# Introduction to Artificial Intelligence
# Semester I, 2025-26

## Bayesian Networks

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# Outline

- Last Class
  - Adversarial Search
- This Class
  - Bayesian Networks
- Reference Material
  - AIMA Ch. 13 and 14

# Acknowledgement

**These slides are intended for teaching purposes only. Some material has been used/adapted from web sources and from slides by Doina Precup, Dorsa Sadigh, Percy Liang, Mausam, Dan Klein, Anca Dragan, Nicholas Roy and others.**

# Uncertainty in AI

- Uncertainty:
  - **Observed variables (evidence)**: Agent knows certain things about the state of the world (e.g., sensor measurements or symptoms)
  - **Unobserved variables**: Agent needs to reason about other aspects (e.g. what disease is present,  is the car operational, location of the burglar)
  - **Model**: Agent knows something about how the known variables relate to the unknown variables
- Probabilistic reasoning gives us a framework for managing our beliefs and knowledge.

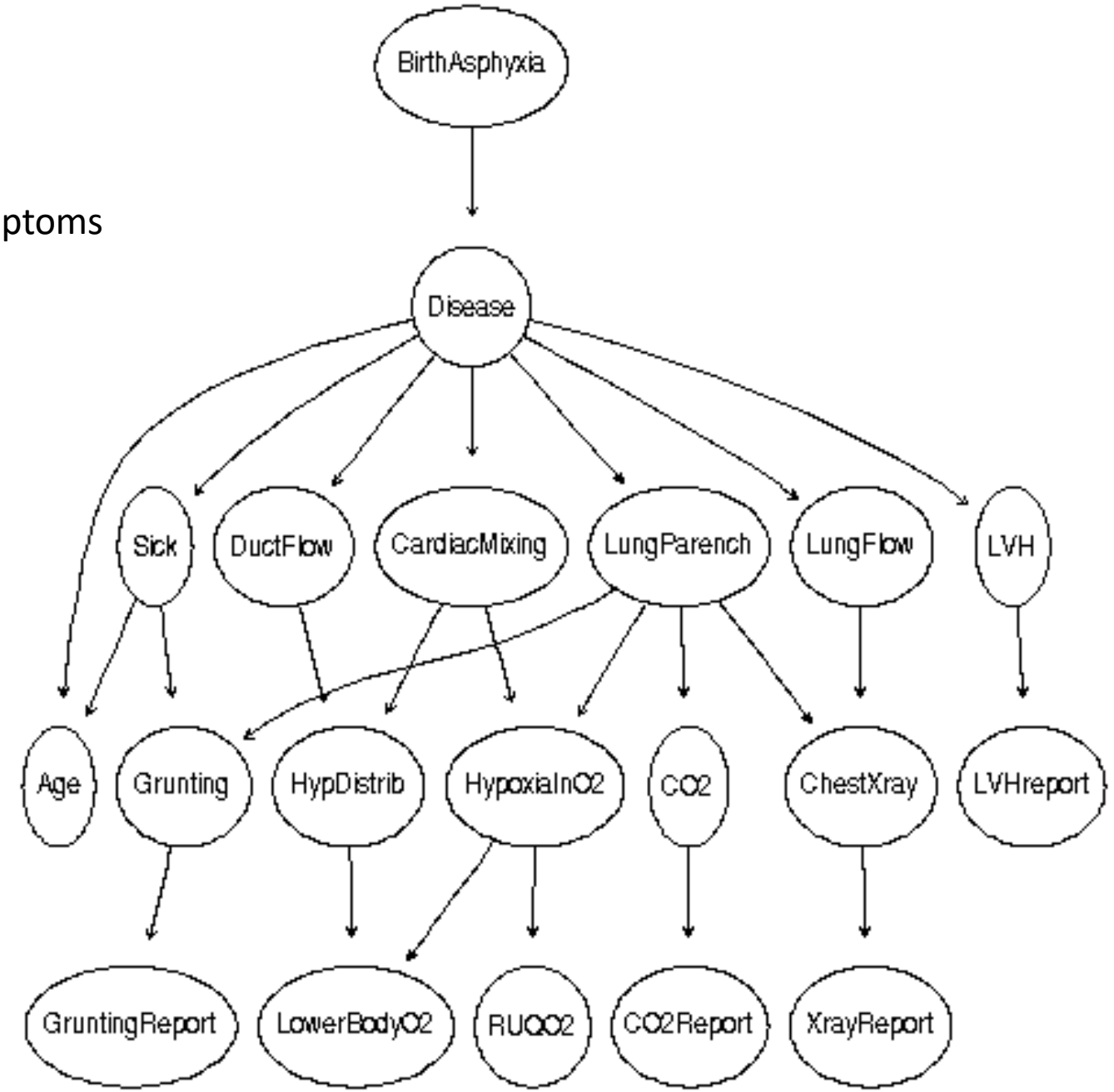
**I hear an unusual sound and a burning smell in my car, what fault is there in my engine?**

**I have fever, loss of smell, loss of taste, do I have Covid?**

**I hear some footsteps in my house, where is the burglar?**

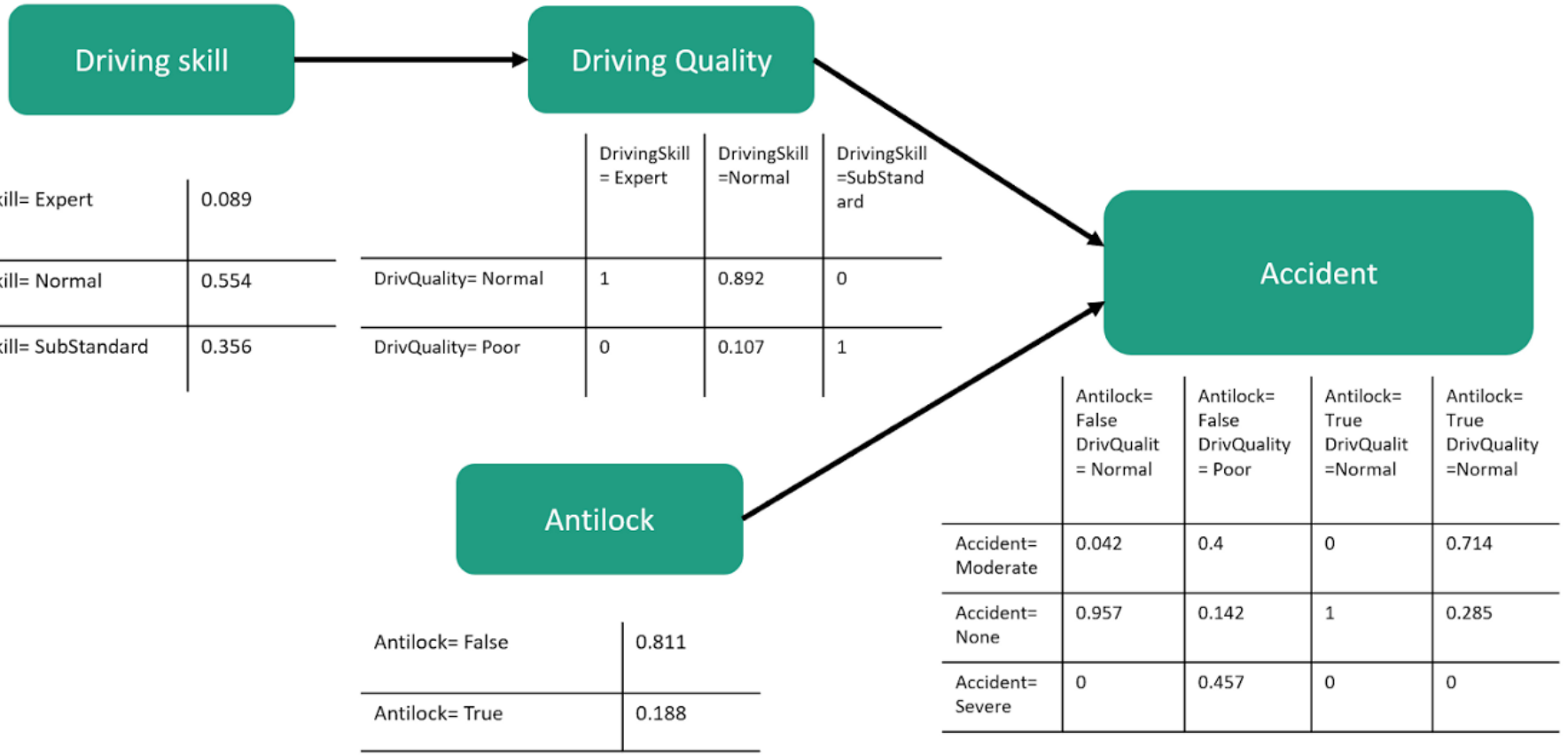
# Examples

Inferring disease from symptoms



# Examples

Accident – driving domain.




| DrivingSkill= Expert | 0.089 |
|---|---|
| DrivingSkill= Normal | 0.554 |
| DrivingSkill= SubStandard | 0.356 |

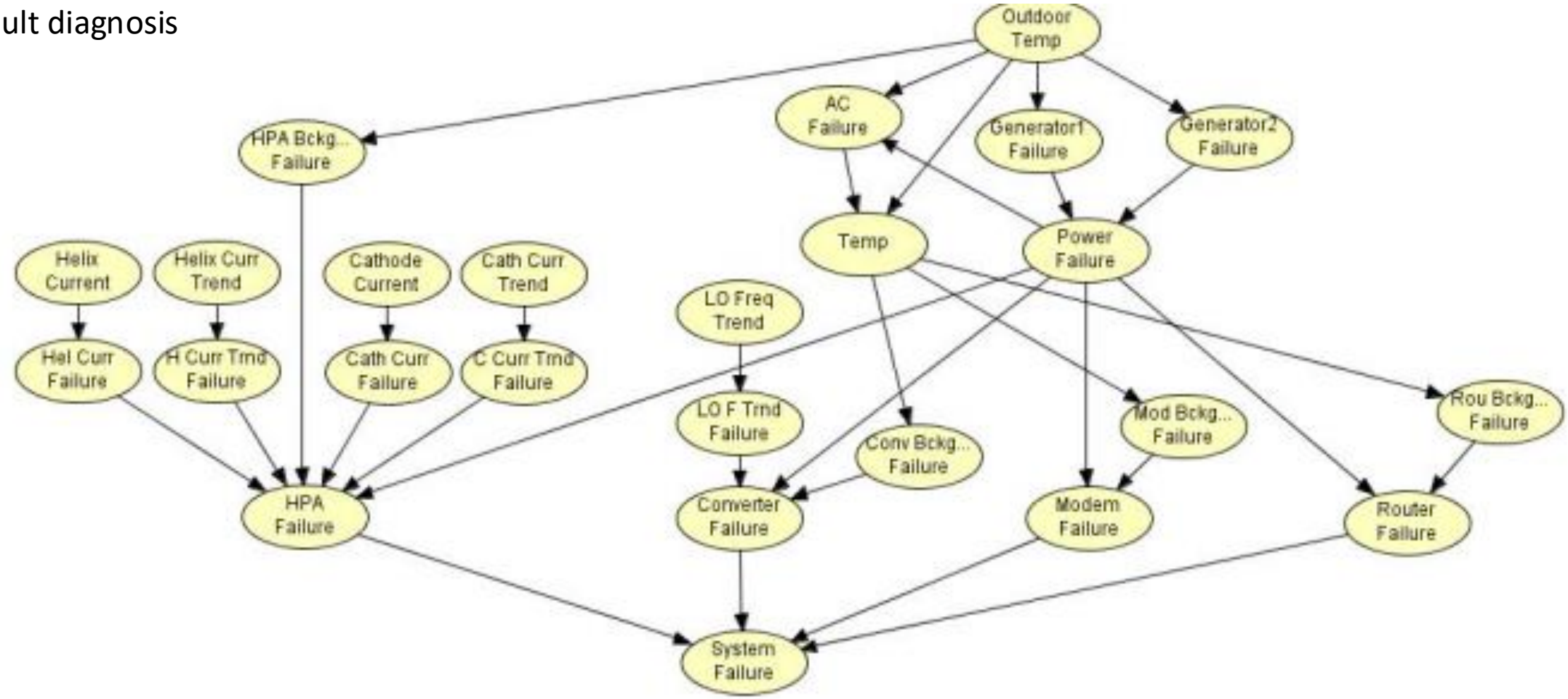
| | DrivingSkill = Expert | DrivingSkill =Normal | DrivingSkill =SubStandard |
|---|---|---|---|
| DrivQuality= Normal | 1 | 0.892 | 0 |
| DrivQuality= Poor | 0 | 0.107 | 1 |

| Antilock= False | 0.811 |
|---|---|
| Antilock= True | 0.188 |

| | Antilock= False DrivQualit = Normal | Antilock= False DrivQuality = Poor | Antilock= True DrivQualit =Normal | Antilock= True DrivQuality =Normal |
|---|---|---|---|---|
| Accident= Moderate | 0.042 | 0.4 | 0 | 0.714 |
| Accident= None | 0.957 | 0.142 | 1 | 0.285 |
| Accident= Severe | 0 | 0.457 | 0 | 0 |

# Examples

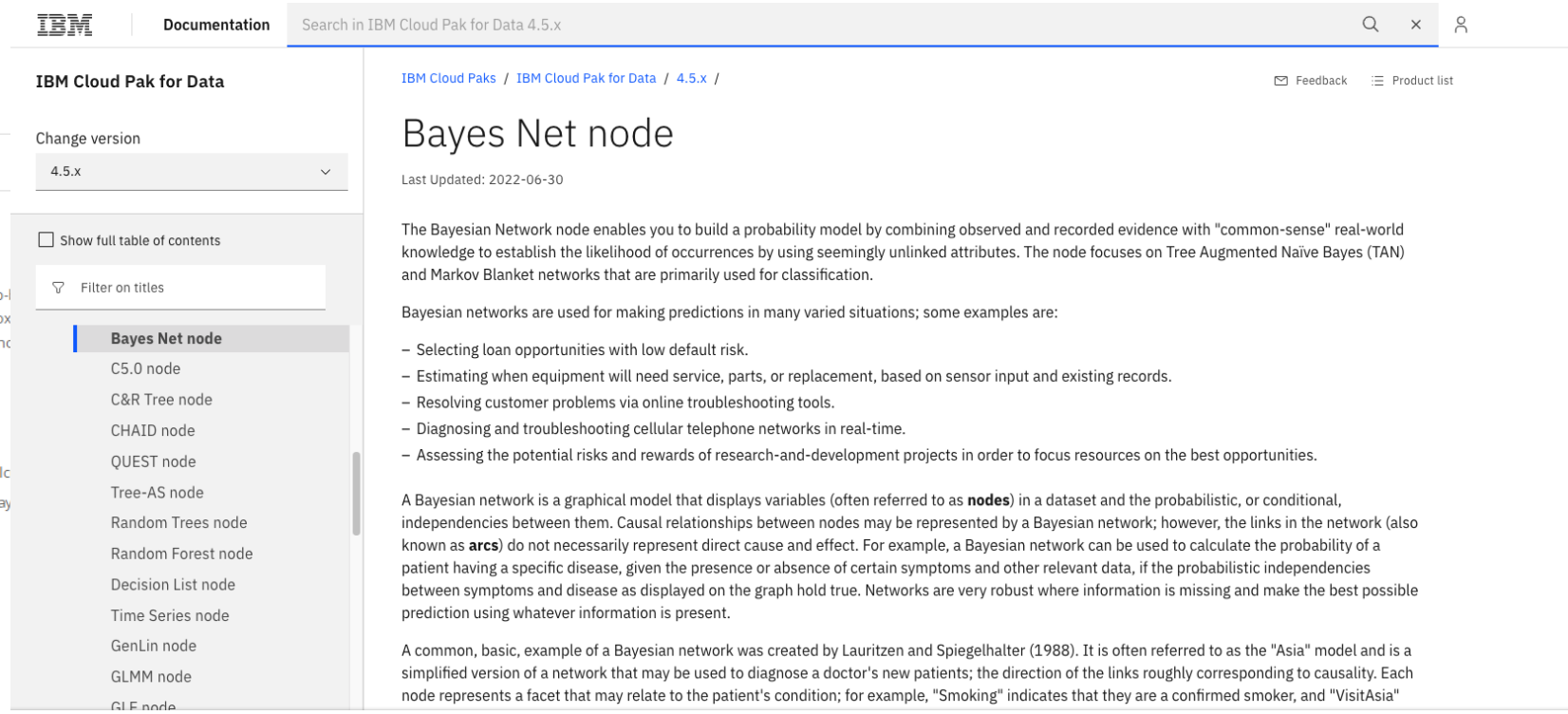
Fault diagnosis



# Examples

Predictive analytics/expert systems



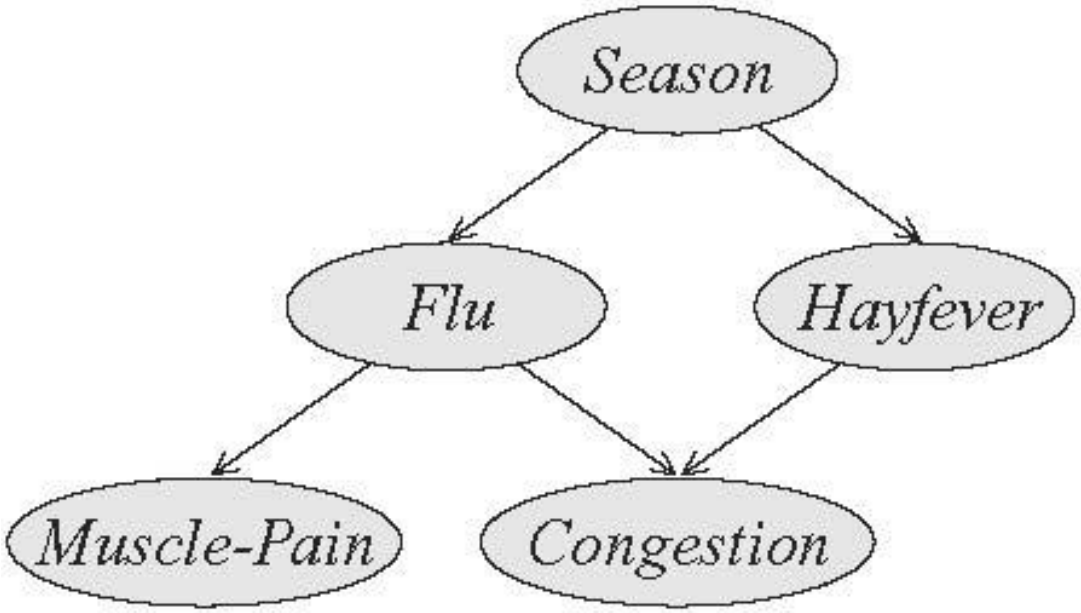



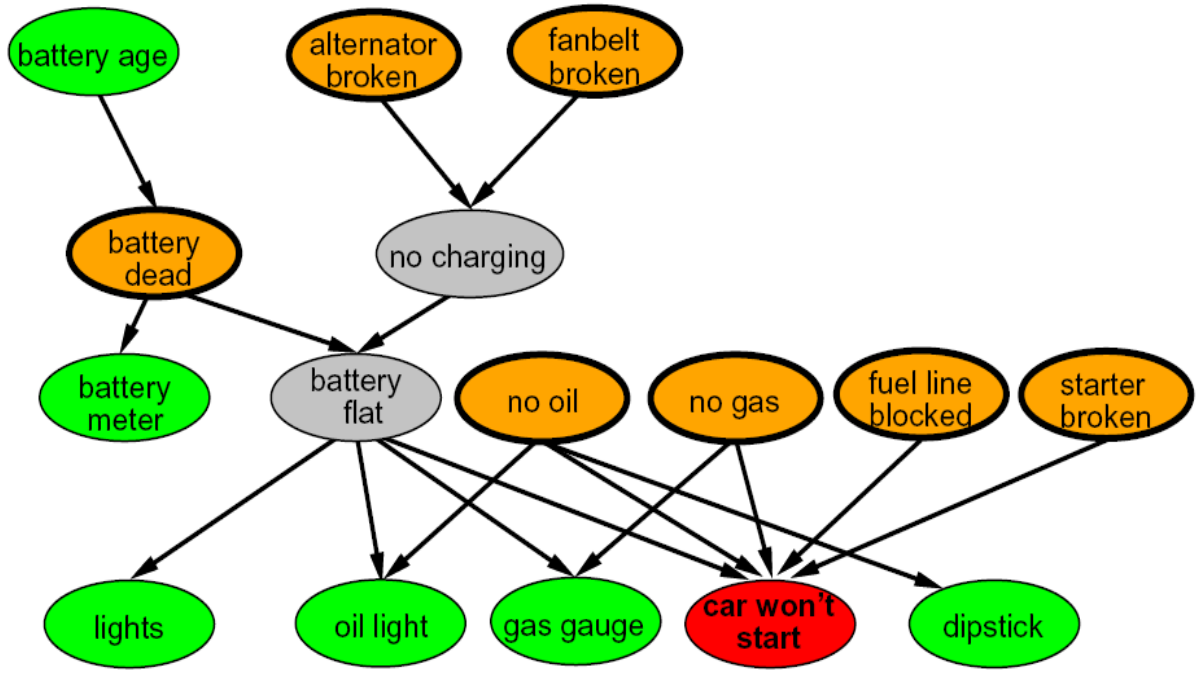
# Bayesian Networks

- Problem with using full joint distribution tables as our probabilistic models:
  - Unless there are only a few variables, the joint is hard to represent explicitly.

- Bayesian Networks:
  - A technique for describing complex joint distributions (models) using simple, local distributions (conditional probabilities)
    - Also known as probabilistic graphical models
  - Encode how variables locally influence each other. Local interactions chain together to give global, indirect interactions

# Examples




Online tool: https://detsutut.shinyapps.io/shinyDBNet/

# Bayesian Networks: Semantics

- A directed, acyclic graph, one node per random variable
- A conditional probability table (CPT) for each node
  - A collection of distributions over X, one for each combination of parents' values

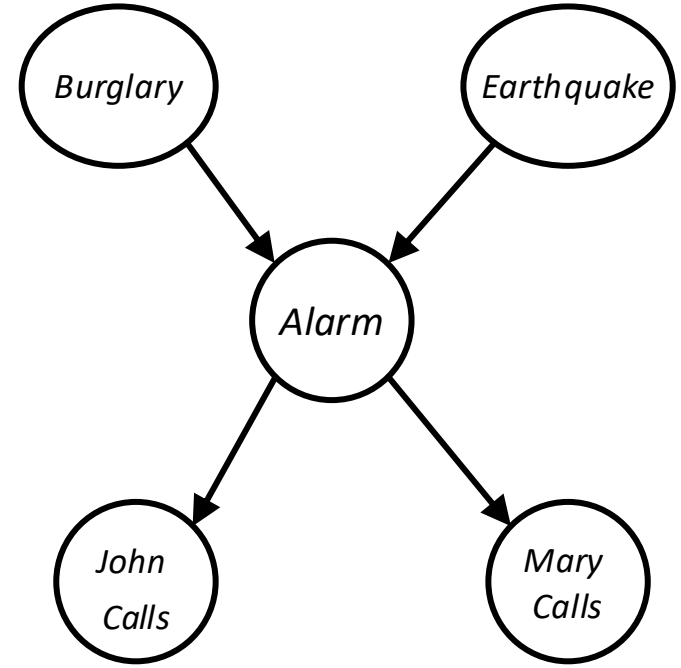
$$P(X|a_1 \ldots a_n)$$

- Bayesian Networks implicitly encode joint distributions
  - As a product of local conditional distributions
  - To see what probability a BN gives to a full assignment, multiply all the relevant conditionals:

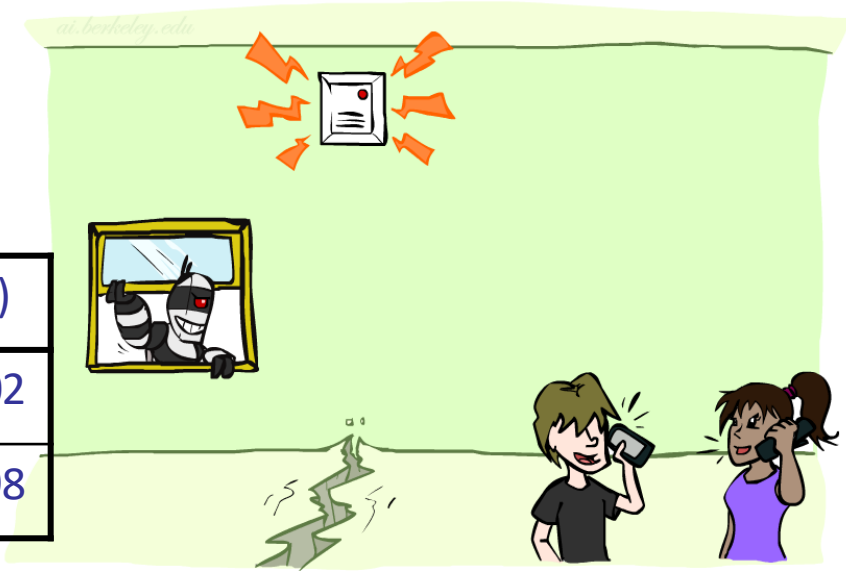
$$P(x_1, x_2, \ldots x_n) = \prod_{i=1}^{n} P(x_i|parents(X_i))$$

# Example: The Alarm Network

| B | P(B) |
|---|---|
| +b | 0.001 |
| -b | 0.999 |




| E | P(E) |
|---|---|
| +e | 0.002 |
| -e | 0.998 |

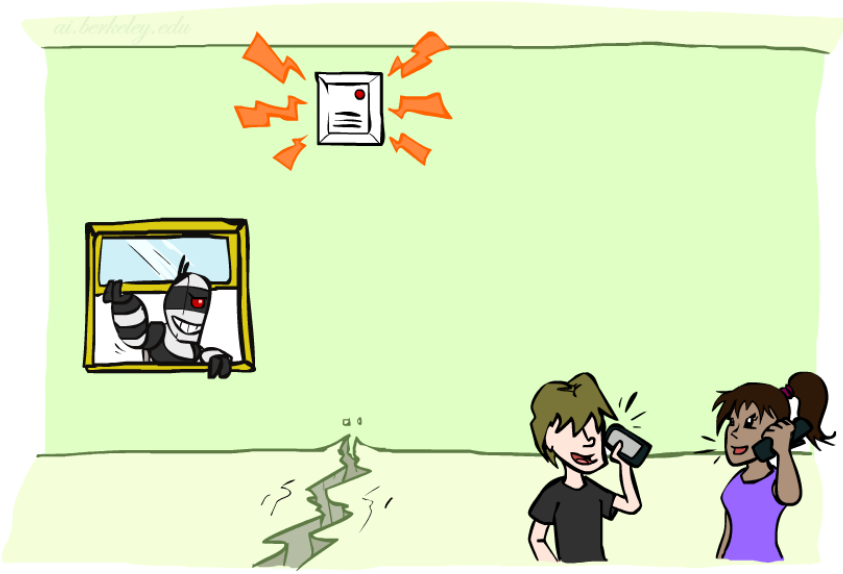


| A | J | P(J\|A) |
|---|---|---|
| +a | +j | 0.9 |
| +a | -j | 0.1 |
| -a | +j | 0.05 |
| -a | -j | 0.95 |

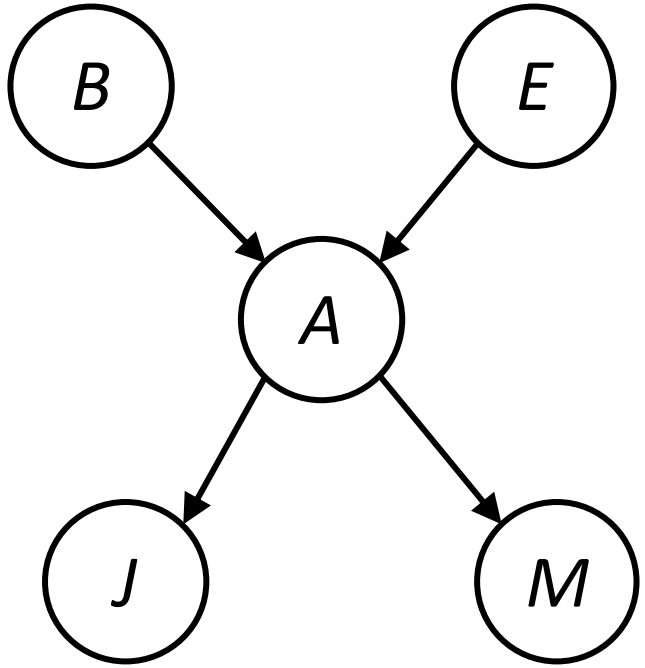
| A | M | P(M\|A) |
|---|---|---|
| +a | +m | 0.7 |
| +a | -m | 0.3 |
| -a | +m | 0.01 |
| -a | -m | 0.99 |

| B | E | A | P(A\|B,E) |
|---|---|---|---|
| +b | +e | +a | 0.95 |
| +b | +e | -a | 0.05 |
| +b | -e | +a | 0.94 |
| +b | -e | -a | 0.06 |
| -b | +e | +a | 0.29 |
| -b | +e | -a | 0.71 |
| -b | -e | +a | 0.001 |
| -b | -e | -a | 0.999 |

# Example: The Alarm Network



| B | P(B) |
|---|---|
| +b | 0.001 |
| -b | 0.999 |




| E | P(E) |
|---|---|
| +e | 0.002 |
| -e | 0.998 |

| A | J | P(J\|A) |
|---|---|---|
| +a | +j | 0.9 |
| +a | -j | 0.1 |
| -a | +j | 0.05 |
| -a | -j | 0.95 |

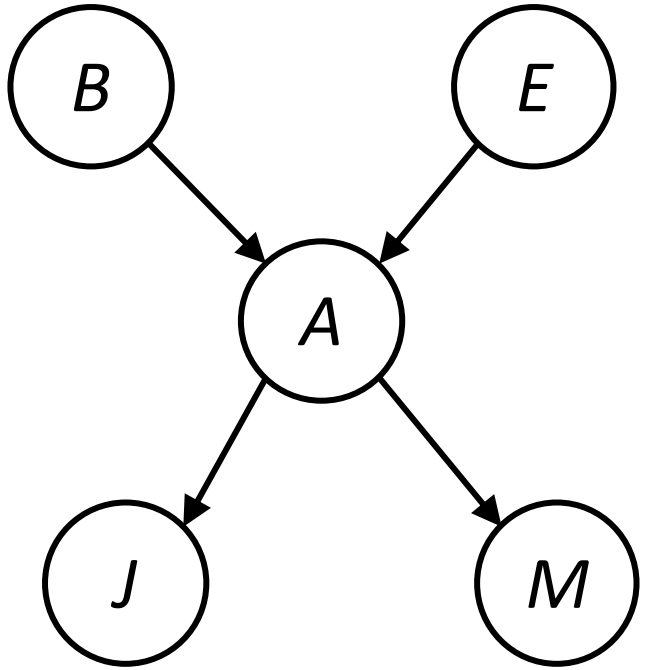
| A | M | P(M\|A) |
|---|---|---|
| +a | +m | 0.7 |
| +a | -m | 0.3 |
| -a | +m | 0.01 |
| -a | -m | 0.99 |

| B | E | A | P(A\|B,E) |
|---|---|---|---|
| +b | +e | +a | 0.95 |
| +b | +e | -a | 0.05 |
| +b | -e | +a | 0.94 |
| +b | -e | -a | 0.06 |
| -b | +e | +a | 0.29 |
| -b | +e | -a | 0.71 |
| -b | -e | +a | 0.001 |
| -b | -e | -a | 0.999 |

$$P(+b, -e, +a, -j, +m) =$$

# Estimating likelihood of variables

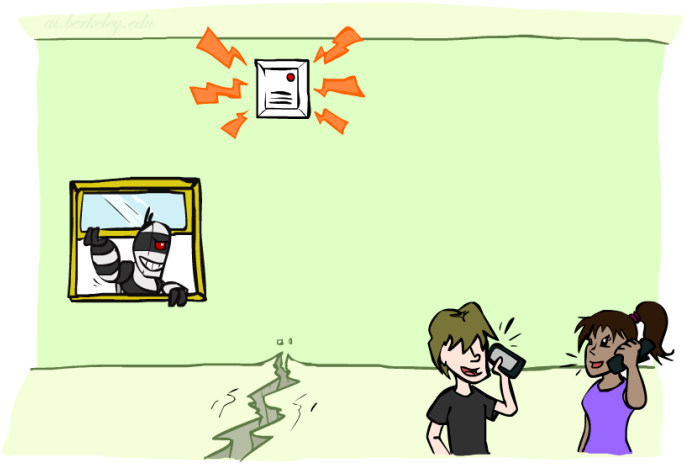
| B | P(B) |
|---|---|
| **+b** | **0.001** |
| -b | 0.999 |




| E | P(E) |
|---|---|
| +e | 0.002 |
| **-e** | **0.998** |

| A | J | P(J\|A) |
|---|---|---|
| +a | +j | 0.9 |
| **+a** | **-j** | **0.1** |
| -a | +j | 0.05 |
| -a | -j | 0.95 |

| A | M | P(M\|A) |
|---|---|---|
| **+a** | **+m** | **0.7** |
| +a | -m | 0.3 |
| -a | +m | 0.01 |
| -a | -m | 0.99 |



| B | E | A | P(A\|B,E) |
|---|---|---|---|
| +b | +e | +a | 0.95 |
| +b | +e | -a | 0.05 |
| **+b** | **-e** | **+a** | **0.94** |
| +b | -e | -a | 0.06 |
| -b | +e | +a | 0.29 |
| -b | +e | -a | 0.71 |
| -b | -e | +a | 0.001 |
| -b | -e | -a | 0.999 |

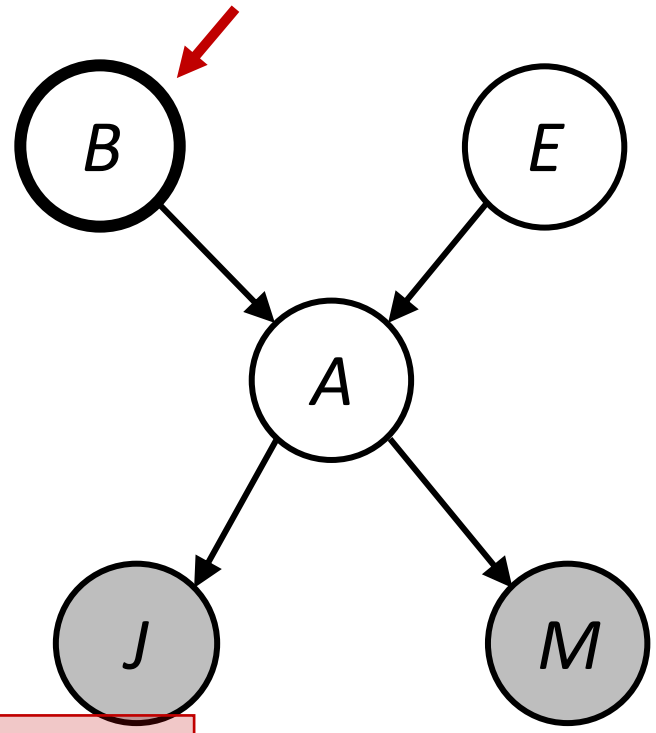
$$P(+b, -e, +a, -j, +m) =$$

$$P(+b)P(-e)P(+a|+b,-e)P(-j|+a)P(+m|+a) =$$

$$0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7$$

# Answering a general probabilistic query

- Inference by enumeration is one way to perform inference in a Bayesian Network (Bayes Net).




$$P(B \mid +j, +m) \propto_B P(B, +j, +m)$$

$$= \sum_{e,a} P(B, e, a, +j, +m)$$

$$= \sum_{e,a} P(B)P(e)P(a|B,e)P(+j|a)P(+m|a)$$

$$= P(B)P(+e)P(+a|B,+e)P(+j|+a)P(+m|+a) + P(B)P(+e)P(-a|B,+e)P(+j|-a)P(+m|-a)$$
$$P(B)P(-e)P(+a|B,-e)P(+j|+a)P(+m|+a) + P(B)P(-e)P(-a|B,-e)P(+j|-a)P(+m|-a)$$

# Bayesian Networks: Inference

- **Bayesian Networks**
  - Implicitly encode a probability distribution
  - As a product of local conditional distributions
- **Variables**
  - Query variables
  - Evidence variables
  - Hidden variables
- **Inference: What we want to estimate?**
  - Estimating some useful quantity from the joint distribution.
  - Posterior probability
  - Most likely explanation

$$P(x_1, x_2, \ldots x_n) = \prod_{i=1}^{n} P(x_i | parents(X_i))$$

$$\left.\begin{array}{l} Q \\ E_1 \ldots E_k = e_1 \ldots e_k \\ H_1 \ldots H_r \end{array}\right\} X_1, X_2, \ldots X_n$$

*All variables*

$$P(Q | E_1 = e_1, \ldots E_k = e_k)$$

$$\text{argmax}_q \; P(Q = q | E_1 = e_1 \ldots)$$

# Inference by Enumeration: A way of answering probabilistic queries

- **Setup:** A distribution over query variables (Q) given evidence variables (E)
  - Select entries consistent with the evidence.
  - E.g., Alarm rang, it is rainy, disease present
- Compute the **joint distribution**
- **Sum out** (eliminate) the hidden variables (H)
- **Normalize** the distribution
- **Next**
  - Introduce a notion called **factors**
  - Understand this computation using joining and marginalization of factors.

$$P(Q|e_1 \dots e_k)$$

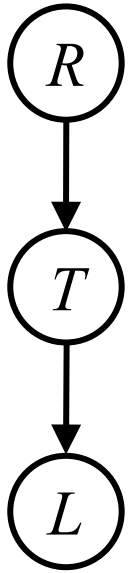
$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} P(Q, h_1 \dots h_r, e_1 \dots e_k)$$

$$Z = \sum_q P(Q, e_1 \cdots e_k)$$

$$P(Q|e_1 \cdots e_k) = \frac{1}{Z} P(Q, e_1 \cdots e_k)$$

# Inference by Enumeration: Example

- **Traffic Domain**
  - Random Variables
    - R: Raining
    - T: Traffic
    - L: Late for class




$$P(L) = ?$$

$$= \sum_{r,t} P(r,t,L)$$

$$= \sum_{r,t} P(r)P(t|r)P(L|t)$$

$P(R)$

| | |
|---|---|
| +r | 0.1 |
| -r | 0.9 |

$P(T|R)$

| | | |
|---|---|---|
| +r | +t | 0.8 |
| +r | -t | 0.2 |
| -r | +t | 0.1 |
| -r | -t | 0.9 |

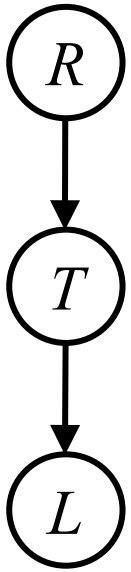
$P(L|T)$

| | | |
|---|---|---|
| +t | +l | 0.3 |
| +t | -l | 0.7 |
| -t | +l | 0.1 |
| -t | -l | 0.9 |

# Inference by Enumeration as Operations on Factors

- **Factors**
  - A factor is a function from some set of variables into a specific value.
  - Initial factos
    - Conditional probability tables (one per node)
    - Select the values consistent with the evidence
- **Inference by Enumeration**
  - Via factors, can be understood as a procedure that joins all the factors and then sums out all the hidden variables.
  - *Define two operations "joining" and "summing" next.*

Traffic domain




$P(R)$

| | |
|---|---|
| +r | 0.1 |
| -r | 0.9 |

$P(T|R)$

| | | |
|---|---|---|
| +r | +t | 0.8 |
| +r | -t | 0.2 |
| -r | +t | 0.1 |
| -r | -t | 0.9 |

$P(L|T)$

| | | |
|---|---|---|
| +t | +l | 0.3 |
| +t | -l | 0.7 |
| -t | +l | 0.1 |
| -t | -l | 0.9 |

If some variables are observed then apply that information to associated factor. Others are not affected.

$L = +\ell$

$P(R)$

| | |
|---|---|
| +r | 0.1 |
| -r | 0.9 |

$P(T|R)$

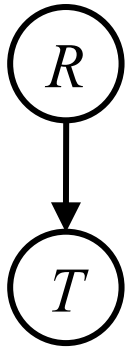
| | | |
|---|---|---|
| +r | +t | 0.8 |
| +r | -t | 0.2 |
| -r | +t | 0.1 |
| -r | -t | 0.9 |

$P(+\ell|T)$

| | | |
|---|---|---|
| +t | +l | 0.3 |
| -t | +l | 0.1 |

# Operation I: Joining Factors

- **Joining**
  - Get all the factors over the joining variables.
  - Build a new factor over the union of variables involved.
  - *Computation for each entry: pointwise products*




$$P(R) \times P(T|R) \Rightarrow P(R,T)$$

| +r | 0.1 |
|---|---|
| -r | 0.9 |

| +r | +t | 0.8 |
|---|---|---|
| +r | -t | 0.2 |
| -r | +t | 0.1 |
| -r | -t | 0.9 |

| +r | +t | 0.08 |
|---|---|---|
| +r | -t | 0.02 |
| -r | +t | 0.09 |
| -r | -t | 0.81 |




$$\forall r, t: \quad P(r,t) = P(r) \cdot P(t|r)$$

# Joining Factors

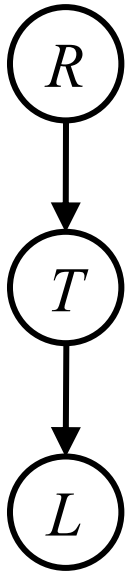
| $A$ | $B$ | $\mathbf{f}_1(A,B)$ | $B$ | $C$ | $\mathbf{f}_2(B,C)$ | $A$ | $B$ | $C$ | $\mathbf{f}_3(A,B,C)$ |
|---|---|---|---|---|---|---|---|---|---|
| T | T | .3 | T | T | .2 | T | T | T | $.3 \times .2 = .06$ |
| T | F | .7 | T | F | .8 | T | T | F | $.3 \times .8 = .24$ |
| F | T | .9 | F | T | .6 | T | F | T | $.7 \times .6 = .42$ |
| F | F | .1 | F | F | .4 | T | F | F | $.7 \times .4 = .28$ |
| | | | | | | F | T | T | $.9 \times .2 = .18$ |
| | | | | | | F | T | F | $.9 \times .8 = .72$ |
| | | | | | | F | F | T | $.1 \times .6 = .06$ |
| | | | | | | F | F | F | $.1 \times .4 = .04$ |

**Figure 14.10** Illustrating pointwise multiplication: $\mathbf{f}_1(A,B) \times \mathbf{f}_2(B,C) = \mathbf{f}_3(A,B,C)$.

$$\mathbf{f}(X_1 \ldots X_j, Y_1 \ldots Y_k, Z_1 \ldots Z_l) = \mathbf{f}_1(X_1 \ldots X_j, Y_1 \ldots Y_k)\, \mathbf{f}_2(Y_1 \ldots Y_k, Z_, \ldots Z_l).$$

Source: AIMA Ch 14.

# Joining Multiple Factors




$P(R)$

| | |
|---|---|
| +r | 0.1 |
| -r | 0.9 |

$P(T|R)$

| | | |
|---|---|---|
| +r | +t | 0.8 |
| +r | -t | 0.2 |
| -r | +t | 0.1 |
| -r | -t | 0.9 |

$P(L|T)$

| | | |
|---|---|---|
| +t | +l | 0.3 |
| +t | -l | 0.7 |
| -t | +l | 0.1 |
| -t | -l | 0.9 |

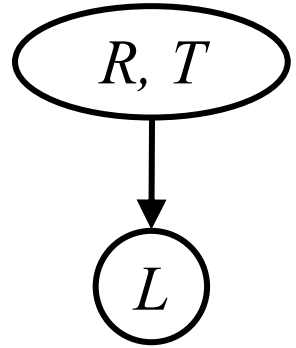
**Join R**

$P(R,T)$

| | | |
|---|---|---|
| +r | +t | 0.08 |
| +r | -t | 0.02 |
| -r | +t | 0.09 |
| -r | -t | 0.81 |

$P(L|T)$

| | | |
|---|---|---|
| +t | +l | 0.3 |
| +t | -l | 0.7 |
| -t | +l | 0.1 |
| -t | -l | 0.9 |




**Join T**




$P(R,T,L)$

| | | | |
|---|---|---|---|
| +r | +t | +l | 0.024 |
| +r | +t | -l | 0.056 |
| +r | -t | +l | 0.002 |
| +r | -t | -l | 0.018 |
| -r | +t | +l | 0.027 |
| -r | +t | -l | 0.063 |
| -r | -t | +l | 0.081 |
| -r | -t | -l | 0.729 |

# Operation II: Eliminating Factors

- **Marginalization**
  - Take a factor and sum out a variable
  - Shrinks the factor to a smaller one

$P(R,T)$

| | | |
|---|---|---|
| +r | +t | 0.08 |
| +r | -t | 0.02 |
| -r | +t | 0.09 |
| -r | -t | 0.81 |

Sum out R

$P(T)$

| | |
|---|---|
| +t | 0.17 |
| -t | 0.83 |

$P(R,T,L)$

$R, T, L$

| | | | |
|---|---|---|---|
| +r | +t | +l | 0.024 |
| +r | +t | -l | 0.056 |
| +r | -t | +l | 0.002 |
| +r | -t | -l | 0.018 |
| -r | +t | +l | 0.027 |
| -r | +t | -l | 0.063 |
| -r | -t | +l | 0.081 |
| -r | -t | -l | 0.729 |

Sum out R

$P(T,L)$

$T, L$

| | | |
|---|---|---|
| +t | +l | 0.051 |
| +t | -l | 0.119 |
| -t | +l | 0.083 |
| -t | -l | 0.747 |

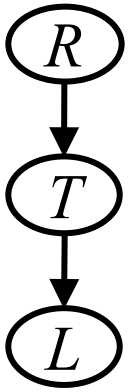
Sum out T

$P(L)$

$L$

| | |
|---|---|
| +l | 0.134 |
| -l | 0.866 |

# Inference by Enumeration




$P(L) = ?$

**Multiple join operations and multiple eliminate operations**

$P(R, T, L)$

$P(L)$

$P(R)$

| | |
|---|---|
| +r | 0.1 |
| -r | 0.9 |

$P(T|R)$

| | | |
|---|---|---|
| +r | +t | 0.8 |
| +r | -t | 0.2 |
| -r | +t | 0.1 |
| -r | -t | 0.9 |

$P(L|T)$

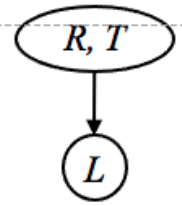
| | | |
|---|---|---|
| +t | +l | 0.3 |
| +t | -l | 0.7 |
| -t | +l | 0.1 |
| -t | -l | 0.9 |

Join R

$P(R, T)$

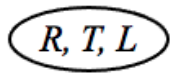
| | | |
|---|---|---|
| +r | +t | 0.08 |
| +r | -t | 0.02 |
| -r | +t | 0.09 |
| -r | -t | 0.81 |

Join T



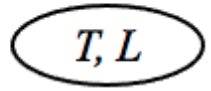

$P(L|T)$

| | | |
|---|---|---|
| +t | +l | 0.3 |
| +t | -l | 0.7 |
| -t | +l | 0.1 |
| -t | -l | 0.9 |




$P(R, T, L)$

| | | | |
|---|---|---|---|
| +r | +t | +l | 0.024 |
| +r | +t | -l | 0.056 |
| +r | -t | +l | 0.002 |
| +r | -t | -l | 0.018 |
| -r | +t | +l | 0.027 |
| -r | +t | -l | 0.063 |
| -r | -t | +l | 0.081 |
| -r | -t | -l | 0.729 |







Sum out R

$P(T, L)$

| | | |
|---|---|---|
| +t | +l | 0.051 |
| +t | -l | 0.119 |
| -t | +l | 0.083 |
| -t | -l | 0.747 |

Sum out T

$P(L)$

| | |
|---|---|
| +l | 0.134 |
| -l | 0.866 |

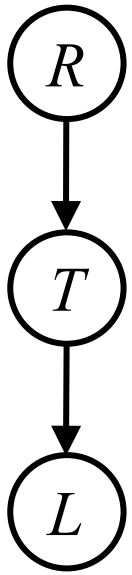
# Variable Elimination

- **Inference by Enumeration**
  - Problem: the whole distribution is “joined up“ before “sum out” the hidden variables

- **Variable Elimination**
  - Interleaves joining and eliminating variables
  - Does not create the full joint distribution in one go
  - *Key Idea:*
    - Picks a variable ordering. Picks a variable.
    - Joins all factors containing that variable.
    - Sums out the influence of the variable on new factor.

  - Leverage the **structure** (topology) of the Bayesian Network
  - Marginalize **early** (avoid growing the full joint distribution)

# Inference by Enumeration vs. Variable Elimination




$$P(L) = ?$$

## Inference by Enumeration

$$= \sum_t \sum_r P(L|t)P(r)P(t|r)$$

Join on r

Join on t

Eliminate r

Eliminate t

## Variable Elimination

$$= \sum_t P(L|t) \sum_r P(r)P(t|r)$$

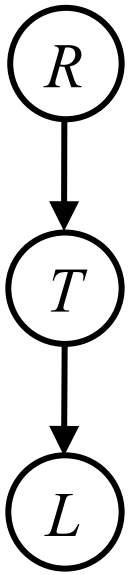
Join on r

Eliminate r

Join on t

Eliminate t

# Variable Elimination




$P(R)$

| | |
|---|---|
| +r | 0.1 |
| -r | 0.9 |

$P(T|R)$

| | | |
|---|---|---|
| +r | +t | 0.8 |
| +r | -t | 0.2 |
| -r | +t | 0.1 |
| -r | -t | 0.9 |

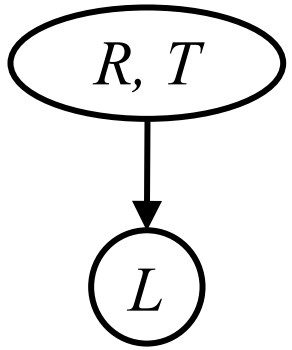
$P(L|T)$

| | | |
|---|---|---|
| +t | +l | 0.3 |
| +t | -l | 0.7 |
| -t | +l | 0.1 |
| -t | -l | 0.9 |

Join R

$P(R,T)$

| | | |
|---|---|---|
| +r | +t | 0.08 |
| +r | -t | 0.02 |
| -r | +t | 0.09 |
| -r | -t | 0.81 |



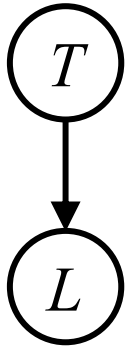

$P(L|T)$

| | | |
|---|---|---|
| +t | +l | 0.3 |
| +t | -l | 0.7 |
| -t | +l | 0.1 |
| -t | -l | 0.9 |

Sum out R

$P(T)$

| | |
|---|---|
| +t | 0.17 |
| -t | 0.83 |




$P(L|T)$

| | | |
|---|---|---|
| +t | +l | 0.3 |
| +t | -l | 0.7 |
| -t | +l | 0.1 |
| -t | -l | 0.9 |

Join T




$P(T,L)$

| | | |
|---|---|---|
| +t | +l | 0.051 |
| +t | -l | 0.119 |
| -t | +l | 0.083 |
| -t | -l | 0.747 |

Sum out T




$P(L)$

| | |
|---|---|
| +l | 0.134 |
| -l | 0.866 |

# Incorporating Evidence

- Till Now, we computed **P(Late)**?
- What happens when **P(Late| Rain)**?
- How to incorporate *evidence* in Variable Elimination.
- **Solution**
  - If evidence, then start with factors and select the evidence.
  - After selecting evidence, eliminate all variables other than query and evidence.

$P(R)$

| | |
|---|---|
| +r | 0.1 |
| -r | 0.9 |

$P(T|R)$

| | | |
|---|---|---|
| +r | +t | 0.8 |
| +r | -t | 0.2 |
| -r | +t | 0.1 |
| -r | -t | 0.9 |

$P(L|T)$

| | | |
|---|---|---|
| +t | +l | 0.3 |
| +t | -l | 0.7 |
| -t | +l | 0.1 |
| -t | -l | 0.9 |

$P(+r)$

| | |
|---|---|
| +r | 0.1 |

$P(T|+r)$

| | | |
|---|---|---|
| +r | +t | 0.8 |
| +r | -t | 0.2 |

$P(L|T)$

| | | |
|---|---|---|
| +t | +l | 0.3 |
| +t | -l | 0.7 |
| -t | +l | 0.1 |
| -t | -l | 0.9 |



Evidence incorporated in the initial factors

# Variable Elimination (VE)

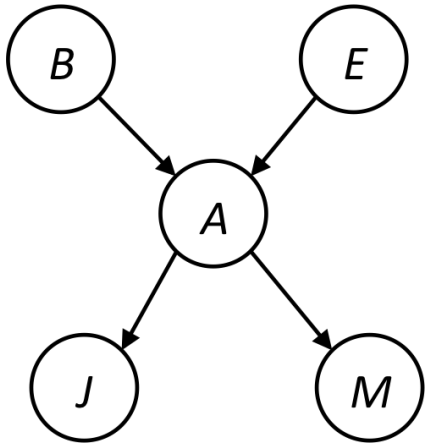
- **Query:** $P(Q|E_1 = e_1, \ldots E_k = e_k)$

- **Start with initial factors:**
  - Local conditional probability tables.
  - Evidence (known) variables are instantiated.

- **While there are still hidden variables (not Q or evidence):**
  - Pick a hidden variable H (from some ordering)
  - Join all factors mentioning H
  - Eliminate (sum out) H

- **Join all the remaining factors and normalize**

# Variable Elimination: Alarm Domain



Model

**Original query** $P(B|j,m) \propto P(B,j,m)$

**Factors**

| | | | | |
|---|---|---|---|---|
| $P(B)$ | $P(E)$ | $P(A\|B,E)$ | $P(j\|A)$ | $P(m\|A)$ |

$$P(B|j,m) \propto P(B,j,m)$$

$$= \sum_{e,a} P(B,j,m,e,a)$$

$$= \sum_{e,a} P(B)P(e)P(a|B,e)P(j|a)P(m|a)$$

$$= \sum_{e} P(B)P(e) \sum_{a} P(a|B,e)P(j|a)P(m|a)$$

$$= \sum_{e} P(B)P(e)f_1(j,m|B,e)$$

$$= P(B) \sum_{e} P(e)f_1(j,m|B,e)$$

$$= P(B)f_2(j,m|B)$$

marginal can be obtained from joint by summing out

use Bayes' net joint distribution expression

use x*(y+z) = xy + xz

joining on a, and then summing out gives $f_1$

use x*(y+z) = xy + xz

joining on e, and then summing out gives $f_2$

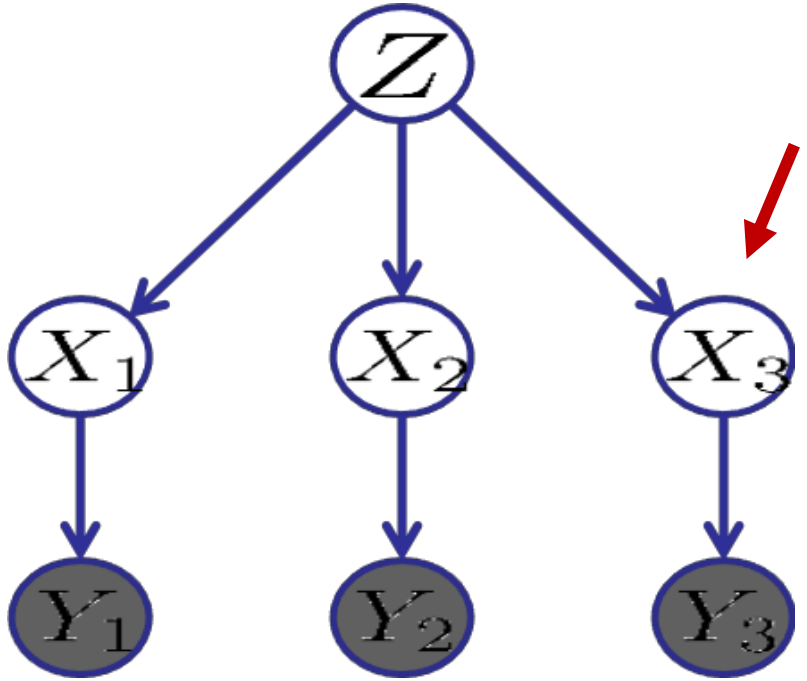
# Variable Elimination: Complexity

Query: $P(X_3|Y_1 = y_1, Y_2 = y_2, Y_3 = y_3)$

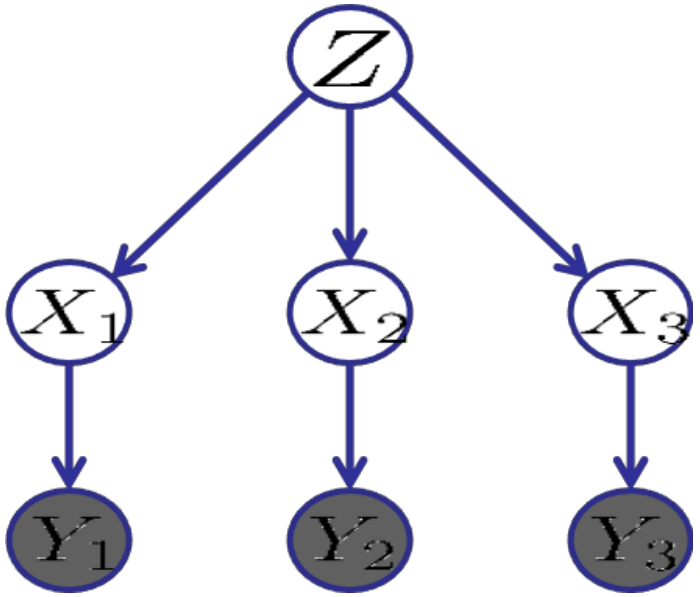
Start by inserting evidence, which gives the following initial factors:

$$P(Z), P(X_1|Z), P(X_2|Z), P(X_3|Z), P(y_1|X_1), P(y_2|X_2), P(y_3|X_3)$$

There are three variables to eliminate { $X_1$, $X_2$ and Z }. The Y variables are observed (instantiated).



# Example




Query: $P(X_3|Y_1 = y_1, Y_2 = y_2, Y_3 = y_3)$

Start by inserting evidence, which gives the following initial factors:

$$P(Z), P(X_1|Z), P(X_2|Z), P(X_3|Z), P(y_1|X_1), P(y_2|X_2), P(y_3|X_3)$$

Eliminate $X_1$, this introduces the factor $f_1(y_1|Z) = \sum_{x_1} P(x_1|Z)P(y_1|x_1)$, and we are left with:

$$P(Z), P(X_2|Z), P(X_3|Z), P(y_2|X_2), P(y_3|X_3), f_1(y_1|Z)$$

Eliminate $X_2$, this introduces the factor $f_2(y_2|Z) = \sum_{x_2} P(x_2|Z)P(y_2|x_2)$, and we are left with:

$$P(Z), P(X_3|Z), P(y_3|X_3), f_1(y_1|Z), f_2(y_2|Z)$$

Eliminate $Z$, this introduces the factor $f_3(y_1, y_2, X_3) = \sum_z P(z)P(X_3|z)f_1(y_1|Z)f_2(y_2|Z)$, and we are left with:

$$P(y_3|X_3), f_3(y_1, y_2, X_3)$$

No hidden variables left. Join the remaining factors to get:

$$f_4(y_1, y_2, y_3, X_3) = P(y_3|X_3), f_3(y_1, y_2, X_3)$$

Normalizing over $X_3$ gives $P(X_3|y_1, y_2, y_3) = f_4(y_1, y_2, y_3, X_3)/\sum_{x_3} f_4(y_1, y_2, y_3, x_3)$

# Variable Elimination: Efficient way to re-use factor computation.

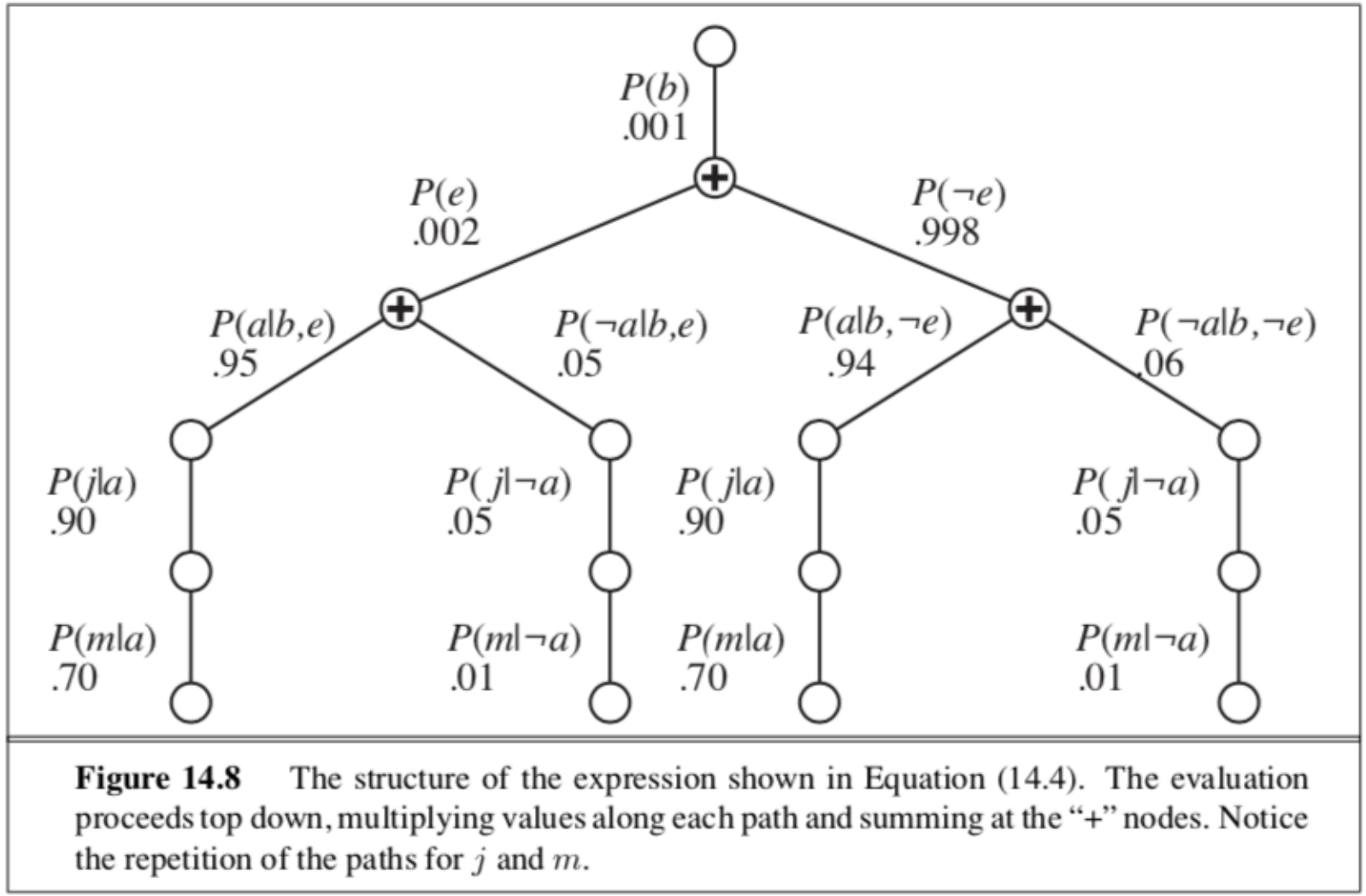



**Figure 14.8** The structure of the expression shown in Equation (14.4). The evaluation proceeds top down, multiplying values along each path and summing at the "+" nodes. Notice the repetition of the paths for $j$ and $m$.

Source: AIMA Ch 14.

# Example

Query: $P(X_3|Y_1 = y_1, Y_2 = y_2, Y_3 = y_3)$

Start by inserting evidence, which gives the following initial factors:

$$P(Z), P(X_1|Z), P(X_2|Z), P(X_3|Z), P(y_1|X_1), P(y_2|X_2), P(y_3|X_3)$$

Eliminate $X_1$, this introduces the factor $f_1(y_1|Z) = \sum_{x_1} P(x_1|Z)P(y_1|x_1)$, and we are left with:

$$P(Z), P(X_2|Z), P(X_3|Z), P(y_2|X_2), P(y_3|X_3), f_1(y_1|Z)$$

Eliminate $X_2$, this introduces the factor $f_2(y_2|Z) = \sum_{x_2} P(x_2|Z)P(y_2|x_2)$, and we are left with:

$$P(Z), P(X_3|Z), P(y_3|X_3), f_1(y_1|Z), f_2(y_2|Z)$$

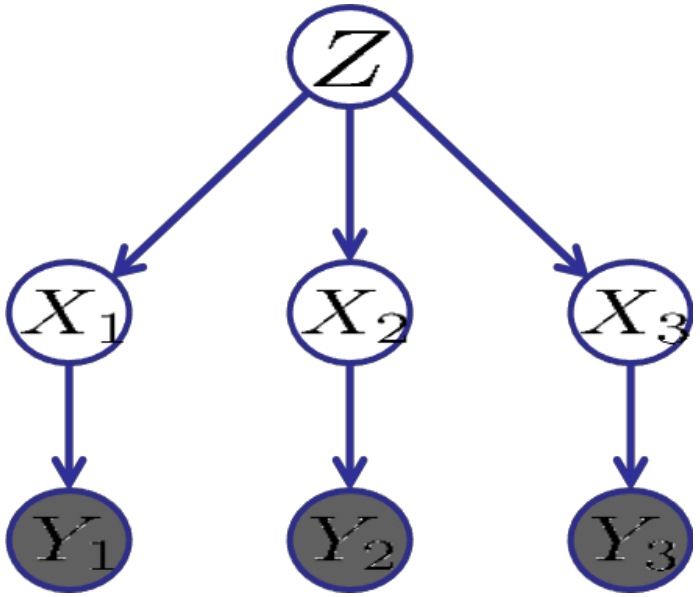
Eliminate $Z$, this introduces the factor $f_3(y_1, y_2, X_3) = \sum_z P(z)P(X_3|z)f_1(y_1|Z)f_2(y_2|Z)$, and we are left with:

$$P(y_3|X_3), f_3(y_1, y_2, X_3)$$

No hidden variables left. Join the remaining factors to get:

$$f_4(y_1, y_2, y_3, X_3) = P(y_3|X_3), f_3(y_1, y_2, X_3)$$

Normalizing over $X_3$ gives $P(X_3|y_1, y_2, y_3) = f_4(y_1, y_2, y_3, X_3)/\sum_{x_3} f_4(y_1, y_2, y_3, x_3)$

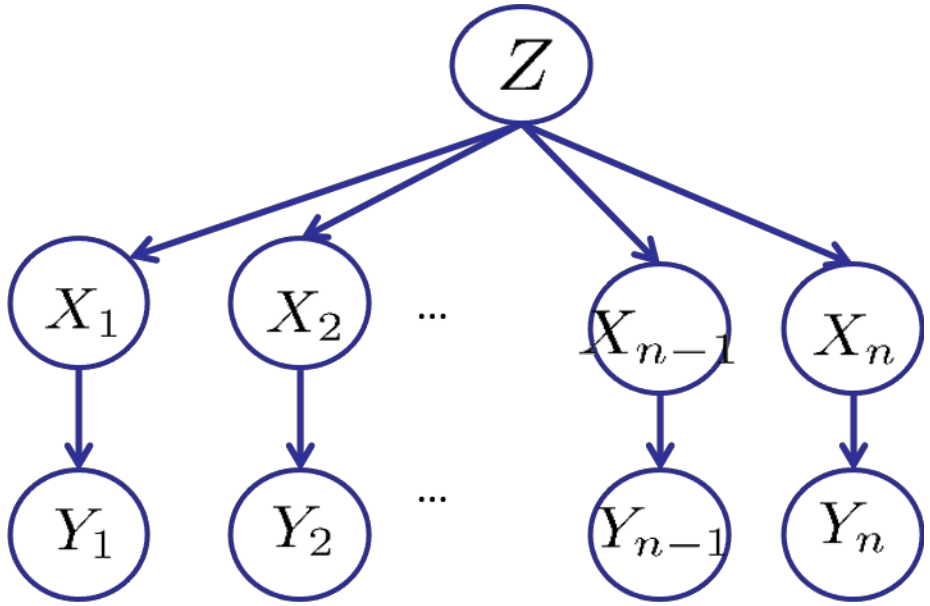



Computational complexity

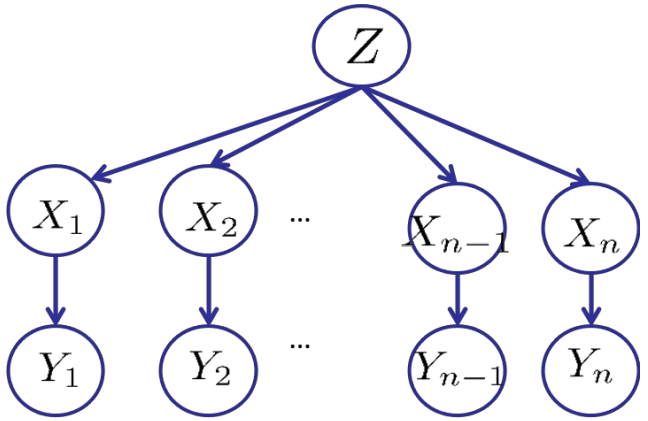
- **Depends on the largest factor generated in VE.**
- Factor size = number of entries in the table.
- In this example: each factor is of size 2 (only one variable). Note that y is observed.
- $X_1$, $X_2$, Z, $X_3$

# How does variable ordering affect VE complexity?

- For the query $P(X_n | y_1,\ldots,y_n)$
- There are n variables Z, $X_1$, …, $X_{n-1}$ to eliminate.
- We need a way to order them (for joining and eliminating)
- Consider two different orderings as
  - Eliminate Z first. Z, $X_1$, …, $X_{n-1}$
  - Eliminate Z last. $X_1$, …, $X_{n-1}$, Z.
  - *What is the size of the maximum factor generated for each of the orderings?*



# Example




**Eliminate Z *First***

$$P(X_n, |y_1, y_2, \ldots, y_n) = \alpha P(Z)P(X_1|Z)P(X_2|Z), \ldots, P(X_n|Z)P(y_1|X_1)P(y_2|X_2), \ldots, P(y_n|X_n)$$

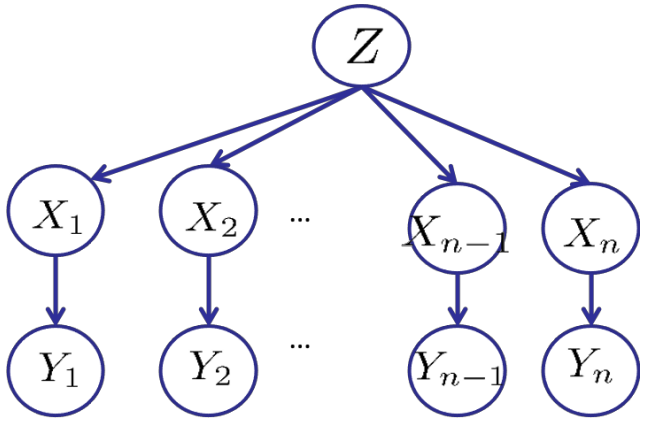
This factor is $2^n$ → $f_1(X_1, X_2, \ldots, X_n) = \sum_z P(z)P(X_1|z), P(X_1|z), \ldots, P(X_1|z)$

$$P(X_n, |y_1, y_2, \ldots, y_n) = \alpha f_1(X_1, X_2, \ldots, X_n)P(y_1|X_1)P(y_2|X_2), \ldots, P(y_n|X_n)$$

$$f_2(X_1, X_2, \ldots, X_{n-1}) = \sum_{x_n} f_1(X_1, X_2, \ldots, X_{n-1}, x_n)P(y_n|x_n)$$

$$P(X_n, |y_1, y_2, \ldots, y_n) = \alpha f_2(X_1, X_2, \ldots, X_{n-1})P(y_1|X_1)P(y_2|X_2), \ldots, P(y_{n-1}|X_{n-1})$$

# Example




## Eliminate Z *Last*

$$P(X_n, |y_1, y_2, \ldots, y_n) = \alpha P(Z)P(X_1|Z)P(X_2|Z), \ldots, P(X_n|Z)P(y_1|X_1)P(y_2|X_2), \ldots, P(y_n|X_n)$$

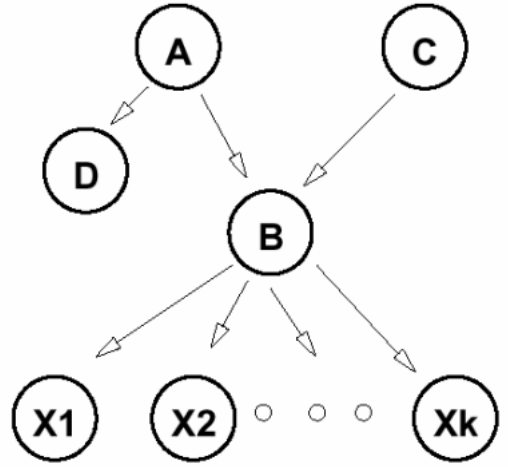
This factor is size 2 → $f_1(y_1|Z) = \sum_{x_1} P(y_1|x_1)P(x_1|Z)$

$$P(X_n, |y_1, y_2, \ldots, y_n) = \alpha P(Z)f_1(y_1|Z)P(X_2|Z), \ldots, P(X_n|Z)P(y_2|X_2), \ldots, P(y_n|X_n)$$

Other steps are like the previous example. Each factor is of size 2 consisting of one variable.
Variable ordering can have considerable impact.

# Variable Ordering for VE

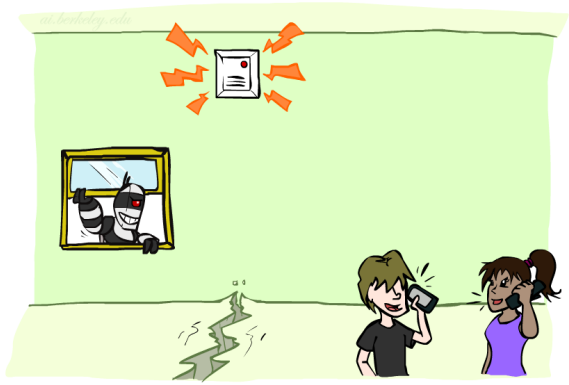
- Variable elimination is dominated by the size of the largest factor constructed during the operation of the algorithm.
- Depends on the structure of the network and order of elimination of the variables.
- Finding the optimal ordering is intractable.
  - Can pose the problem of finding good ordering as a search.
  - Use heuristics.
- Min-fill heuristic
  - Eliminate the variable that creates the smallest sized factor (greedy approach).
- Min-neighbors
  - Eliminate the variable that has the smallest number of neighbors in the current graph.

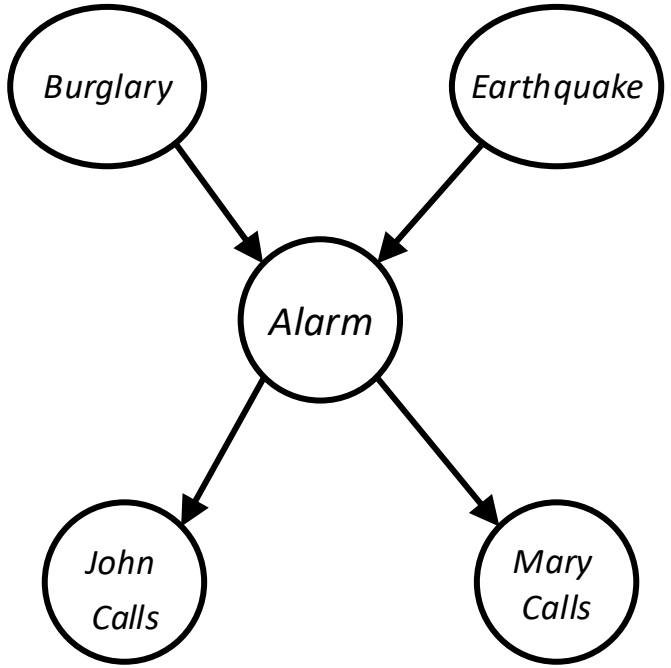



Rank A, B and D with the Min-Fill heuristic.

# Some variables may be irrelevant for VE

**Every variable that is not an ancestor of a query variable or evidence variable is irrelevant for the query.**






$$P(J)$$

$$= \Sigma_{M,A,B,E}\, P(J,M,A,B,E)$$

$$= \Sigma_{M,A,B,E}\, P(J|A)P(B)P(A|B,E)P(E)P(M|A)$$

$$= \Sigma_{A}P(J|A)\, \Sigma_{B}P(B)\, \Sigma_{E}P(A|B,E)P(E)\, \Sigma_{M}P(M|A)$$

**Variable can be eliminated.**



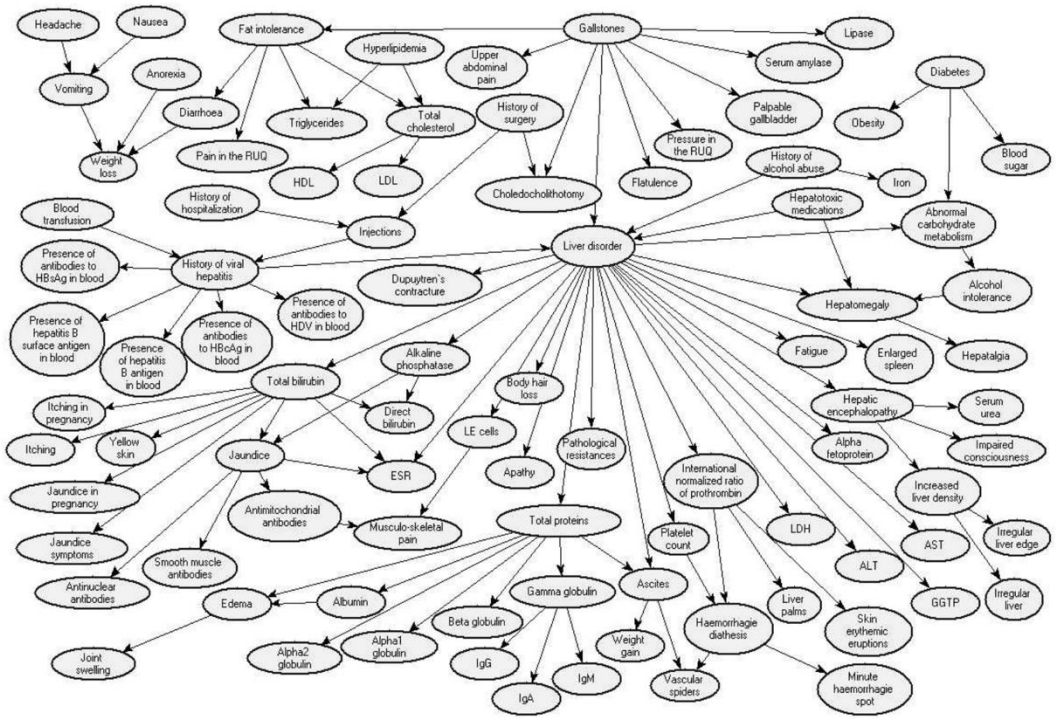
Slide adapted from Prof. Mausam

# Bayesian Networks: Independence

- Bayesian Networks
  - Implicitly encode joint distributions
  - A collection of distributions over X, one for each combination of parents' values
  - Product of local conditional distributions
- Inference
  - Given a fixed BN, what is P(X | e)
  - Variable Elimination
- **Modeling**
  - Understanding the assumptions made when choosing a Bayes net graph

$$P(X|a_1 \dots a_n)$$

$$P(x_1, x_2, \dots x_n) = \prod_{i=1}^{n} P(x_i|parents(X_i))$$




A Bayesian Network Model for Diagnosis of Liver Disorders. Onisko et al. 99.

# Conditional Independence

- X and Y are **independent** if

$$\forall x, y \;\; P(x, y) = P(x)P(y) \;\; \dashrightarrow \;\; X \perp\!\!\!\perp Y$$

- X and Y are **conditionally independent** given Z

$$\forall x, y, z \;\; P(x, y|z) = P(x|z)P(y|z) \dashrightarrow X \perp\!\!\!\perp Y|Z$$

- (Conditional) independence: Given Z, Y has no more information to convey about X or Y does not probabilistically influence X.

- Example: $Alarm \perp\!\!\!\perp Fire|Smoke$

Smoke causes the alarm to be triggered. Once there is smoke it does not matter what caused it (e.g., Fire or any other source).

# Graph structure encodes independence relations

- Conditional independence relations in a Bayes Net

Bayes net: $p(\mathbf{x}) = \prod_{i=1}^{N} p(x_i \mid \mathbf{x}_{\text{pa}(i)})$

Factorization $\iff$ conditional independence (one statement per node)

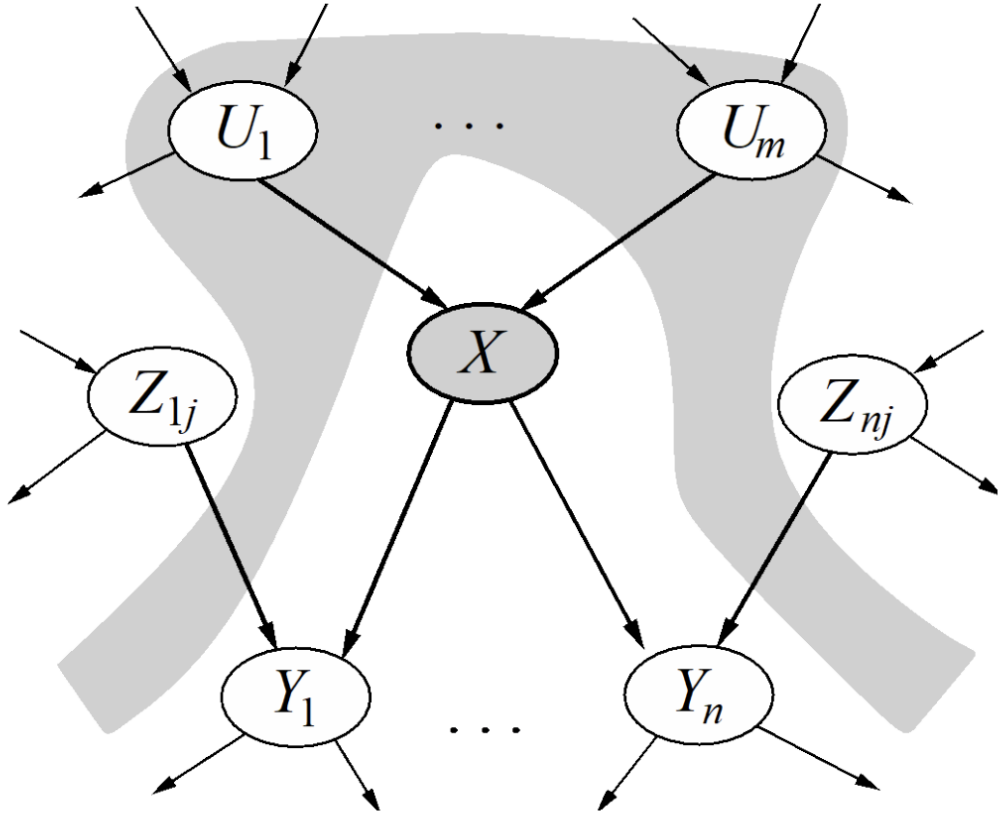
$$p(\mathbf{x}) = \prod_{i=1}^{N} p(x_i \mid \mathbf{x}_{\text{pa}(i)}) \quad \iff \quad X_i \perp \mathbf{X}_{\text{nd}(i)} \mid \mathbf{X}_{\text{pa}(i)} \text{ for all } i$$

factorization $\iff$ conditional independence

RHS in words: $X_i$ **is conditionally independent of its non-descendants given its parents**

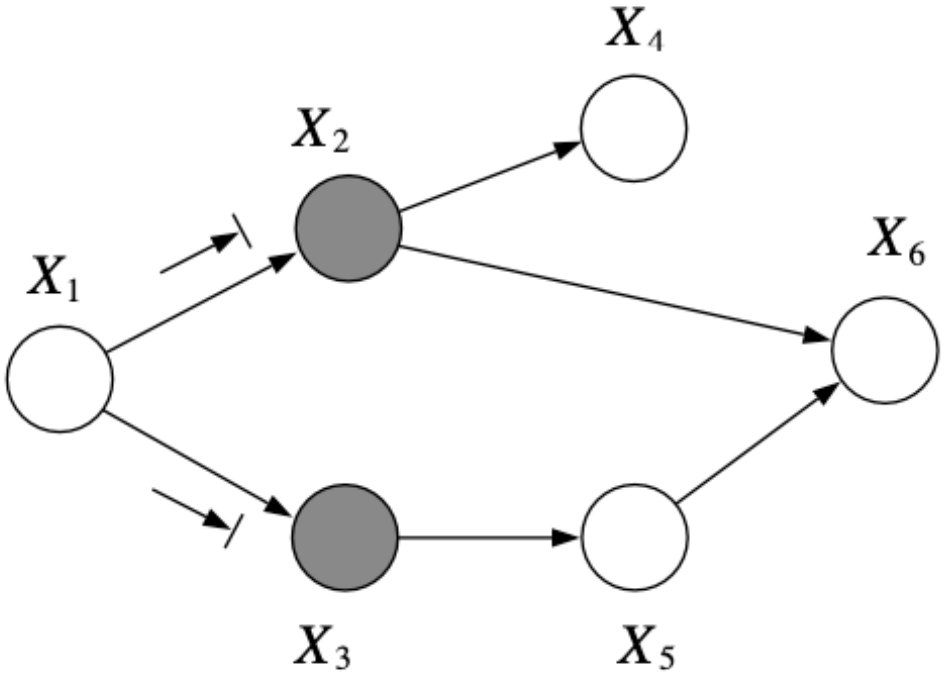
# Graph structure encodes independence relations

**Each node is conditionally independent of all its ancestor nodes (non-descendants) in the graph, given all of its parents.**



# Graph structure encodes independence relations

So far, we know $X_i \perp X_{\text{nd}(i)} | X_{\text{pa}(i)}$ for all $i$




ans is no

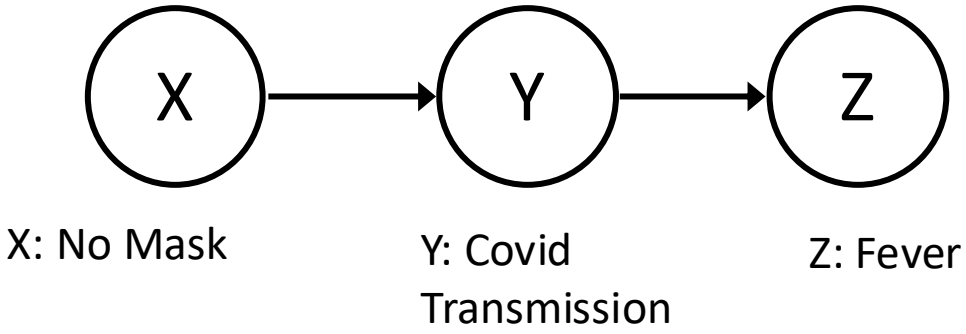
Is X1 conditionally independent of X6 given X3 and X4?

but if X2 and X3 are given than the answer is yes

# Bayesian Network: Independence Assumptions

- Often there are **additional conditional independences** that are implicit in the network.
- Core Idea: examine three node networks and then chain the ideas together.
- How to show if two variables (X and Y) are conditionally independent given evidence (say Z)?
  - **Yes.** Provide a proof by analyzing the probability expression.
  - **No.** Find a counter example. Instantiate a CPT for the BN such that X and Y are not independent given Z.

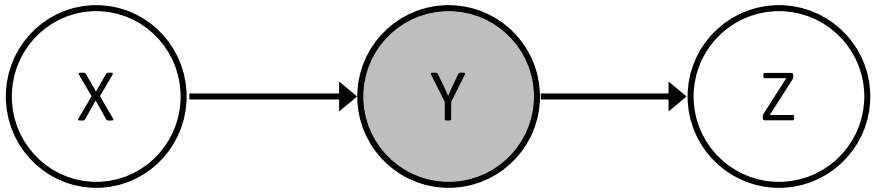
# Causal Chains




- **Is X guaranteed to be independent of Z?**
  - **No**
- Intuitively
  - Wearing no masks causes virus transmission which causes fever.
  - Wearing masks causes no virus transmission causes no symptom.
  - Path between X and Z is active.
- Instantiate a CPT

$$P( +y \mid +x ) = 1, P( -y \mid - x ) = 1,$$
$$P( +z \mid +y ) = 1, P( -z \mid -y ) = 1$$

# Causal Chains




X: No Mask

Y: Covid Transmission

Z: Fever

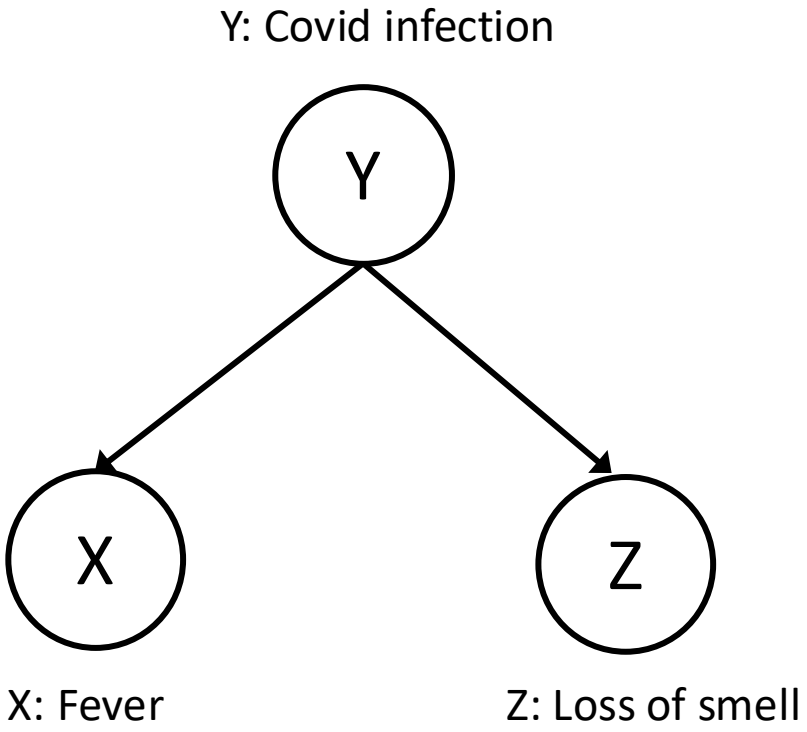
$$P(x, y, z) = P(x)P(y|x)P(z|y)$$

- **Is X guaranteed to be independent of Z given Y?**
  - **Yes**

$$P(z|x, y) = \frac{P(x, y, z)}{P(x, y)}$$
$$= \frac{P(x)P(y|x)P(z|y)}{P(x)P(y|x)}$$
$$= P(z|y)$$

- **Evidence along the chain blocks the influence (inactivates the path).**

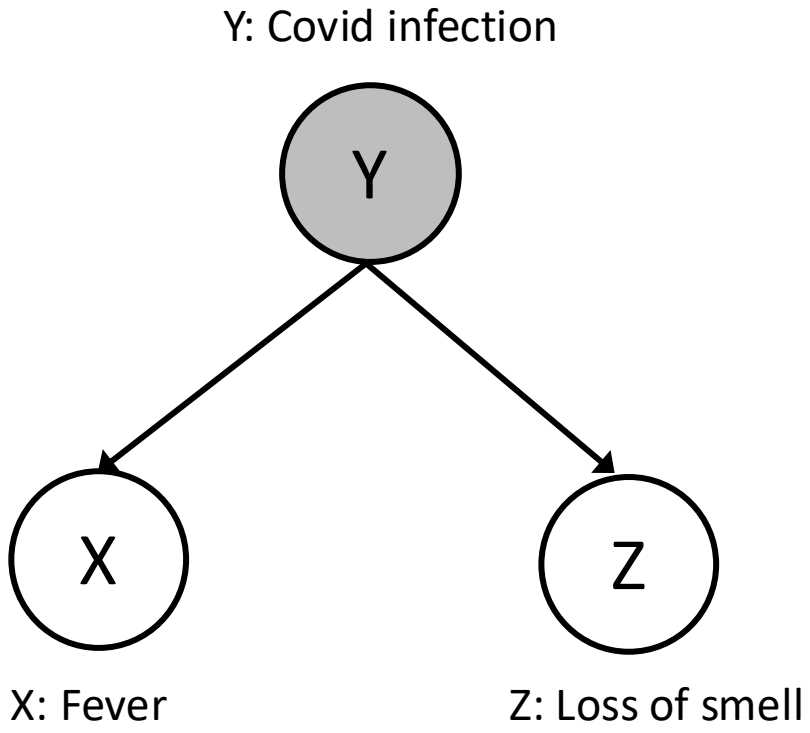
# Common Cause




- **Is X guaranteed to be independent of Z?**
  - **No**
- Intuitively
  - Covid infection causes both Fever and Loss of Smell.
  - Path between X and Z is active.
- Instantiate a CPT

$$P( +x \mid +y ) = 1,\ P( -x \mid -y ) = 1,$$
$$P( +z \mid +y ) = 1,\ P( -z \mid -y ) = 1$$

# Common Cause




$$P(x, y, z) = P(y)P(x|y)P(z|y)$$

- **Is X independent of Z given Y?**
  - **Yes**

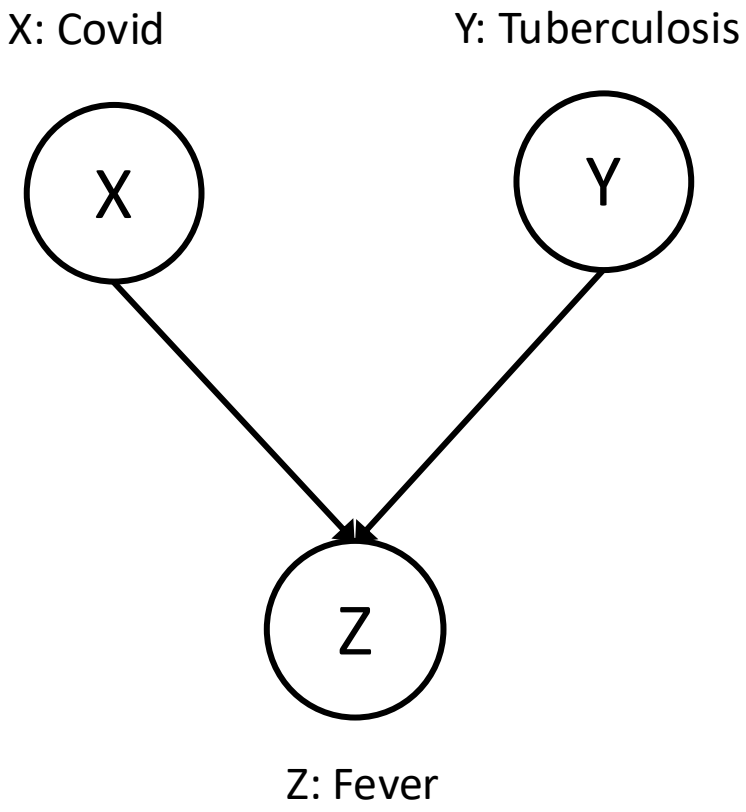
$$P(z|x, y) = \frac{P(x, y, z)}{P(x, y)}$$

$$= \frac{P(y)P(x|y)P(z|y)}{P(y)P(x|y)}$$

$$= P(z|y)$$

- If you have Covid, then belief over the loss of smell is not affected by presence of fever
- **Observing the cause blocks the influence (inactivates the path).**

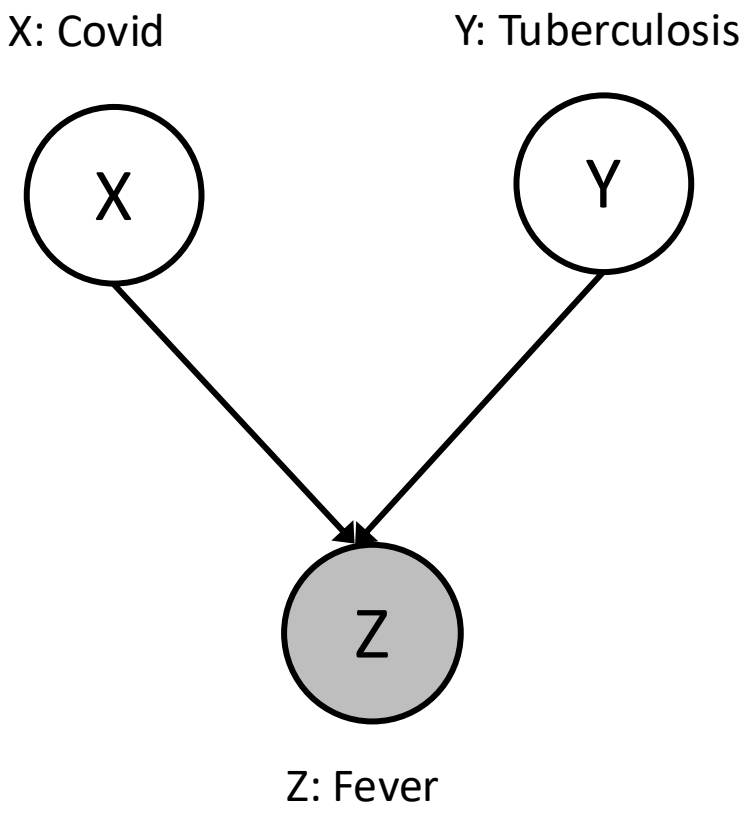
# Common Effect




- **Are X and Y independent?**
  - **Yes**
  - Covid and TB both cause Fever. But can't say that if you have Covid then you are more or less likely to have TB (under this model)

$$
\begin{aligned}
P(x,y) &= \sum_z P(x,y,z) \\
&= \sum_z P(x)P(y)P(z|x,y) \\
&= P(x)P(y)\sum_z P(z|x,y) \\
&= P(x)P(y)
\end{aligned}
$$

# Common Effect




- **Is X independent of Y given Z?**
  - **No**
- Seeing the fever puts Covid and TB in competition as possible causal explanations.
- It is likely that one of them is the cause, rare for both. If Covid is present then the likelihood of TB being present is low (reduces its chances).
- **Observing the cause activates influence between possible causes.**

# "Explaining Away"

Consider the network

```
Battery -> Gauge <- FuelTank
```

Here are some CPTs:

$$\Pr(B = 1) = 0.9$$
$$\Pr(F = 1) = 0.9$$
$$\Pr(G = 1 \mid B = 1, F = 1) = 0.8$$
$$\Pr(G = 1 \mid B = 1, F = 0) = 0.2$$
$$\Pr(G = 1 \mid B = 0, F = 1) = 0.2$$
$$\Pr(G = 1 \mid B = 0, F = 0) = 0.1$$

- What is the prior that the tank is empty? $\Pr(F = 0) = 0.1$
- What if we observe the fuel gauge and find that it reads empty? $\Pr(F = 0 \mid G = 0) \approx 0.257$
- Now, what if we find the battery is dead? $\Pr(F = 0 \mid G = 0, B = 0) \approx 0.111$ The probability that the tank is empty has decreased! Finding that the battery is flat explains away the empty fuel tank reading.



In words: if there are two possible causes for the observed evidence, knowing about one of the causes provides information about the other.

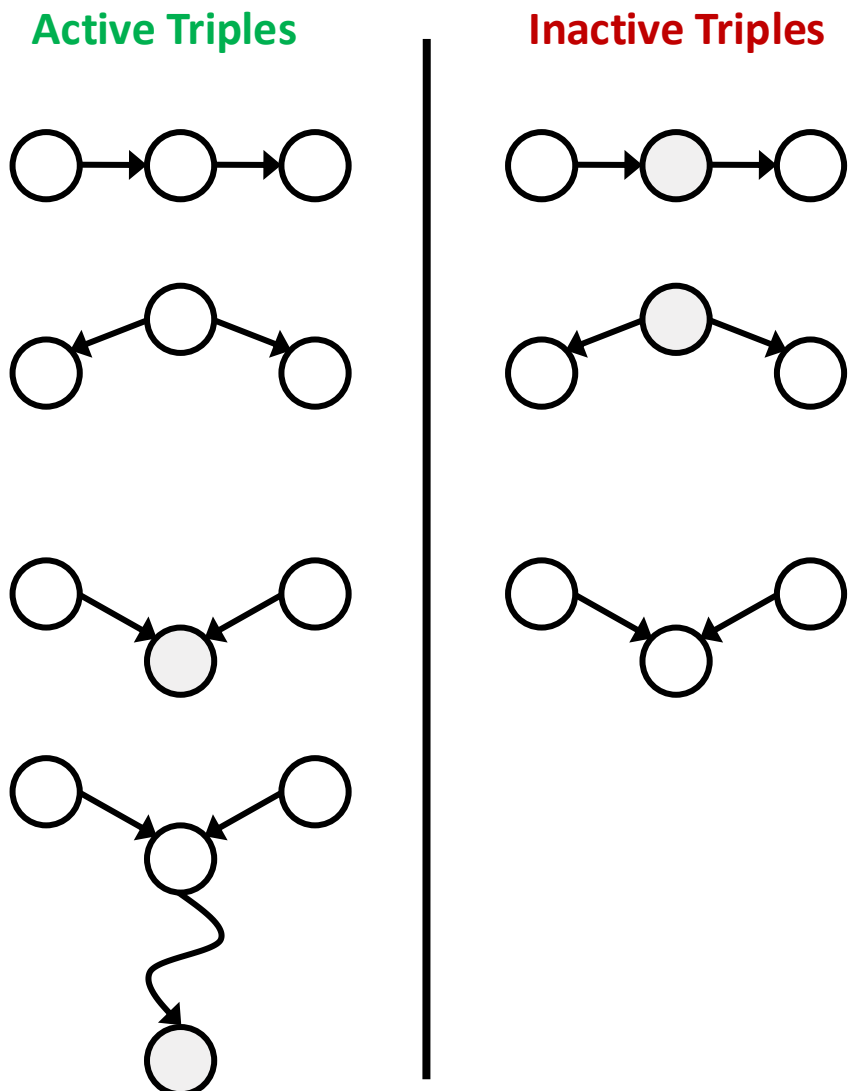
Example from Nicholas Roy

# Active and Inactive Paths

- Question: Are X and Y conditionally independent given evidence variables {Z}?
  - Yes, if X and Y **"d-separated"** by Z
  - Consider all (**undirected**) paths from X to Y
  - No active paths = independence.

- A **path is active** if **each triple is active**:
  - Causal chain A -> B -> C where B is unobserved (either direction)
  - Common cause A <- B -> C where B is unobserved
  - Common effect (aka v-structure)
    A -> B <- C where B *or one of its descendants* is observed

- A path is blocked with even a single inactive segment



# D-Separation

- Query: $X_i \perp\!\!\!\perp X_j | \{X_{k_1}, ..., X_{k_n}\}$ ?
- Check all (undirected) paths between $X_i$ and $X_j$
  - If one or more active, then independence not guaranteed

$$X_i \not\perp\!\!\!\perp X_j | \{X_{k_1}, ..., X_{k_n}\}$$

  - Otherwise (i.e. if all paths are inactive), then independence is guaranteed

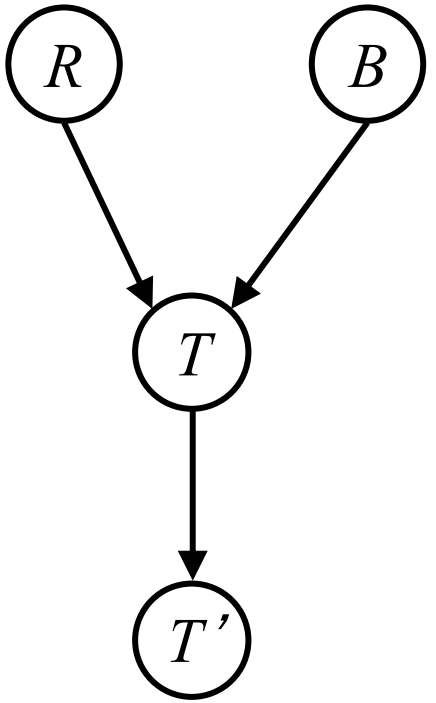
$$X_i \perp\!\!\!\perp X_j | \{X_{k_1}, ..., X_{k_n}\}$$

# D-Separation: Examples

$R \perp\!\!\!\perp B$ *Yes*

$R \perp\!\!\!\perp B | T$

$R \perp\!\!\!\perp B | T'$



# D-Separation: Examples

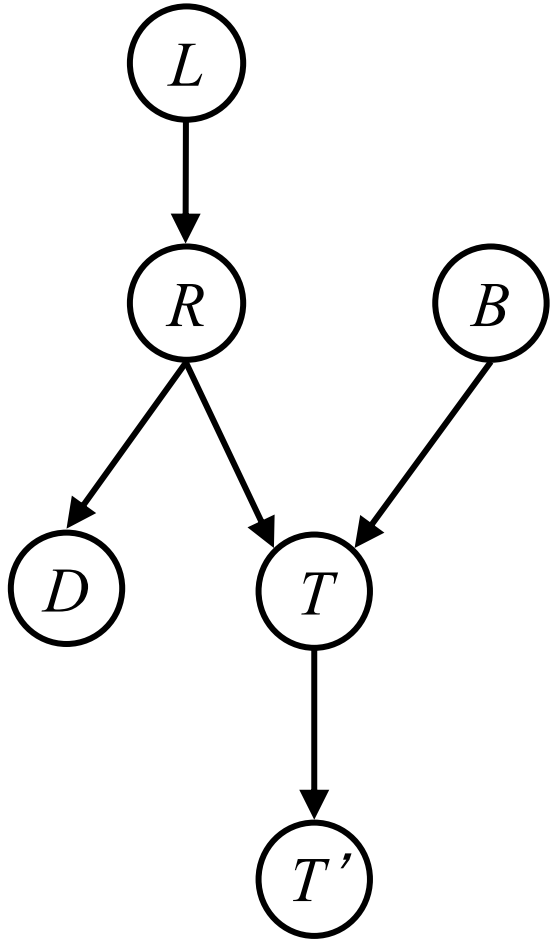
$L \perp\!\!\!\perp T' | T$ *Yes*

$L \perp\!\!\!\perp B$ *Yes*

$L \perp\!\!\!\perp B | T$

$L \perp\!\!\!\perp B | T'$

$L \perp\!\!\!\perp B | T, R$ *Yes*

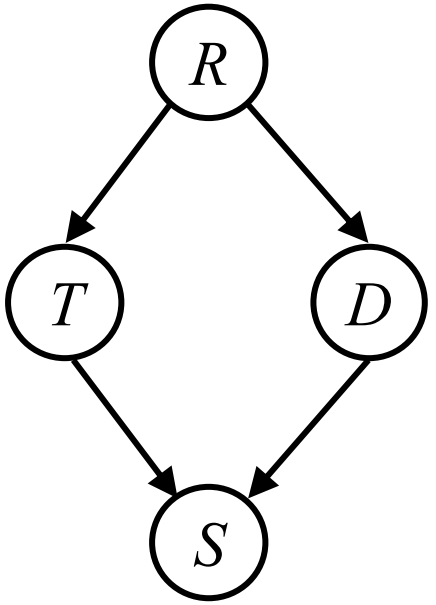


# D-Separation: Examples

$T \perp\!\!\!\perp D$

$T \perp\!\!\!\perp D | R$ *Yes*

$T \perp\!\!\!\perp D | R, S$



# Take aways

- Bayesian Networks
  - Assume that there is uncertainty in the domain. They model aspects of the domain and their relationships as probability distributions.
  - The structure of the network (edges and CPTs) encode how we think the domain works. What aspects influence which other aspects.
  - The structure constrains direct association from naïve association and in turn encodes certain independences that may exist in the domain.
  - Crucially, this influence is probabilistic or imperfect. It is not “if and only if” kind of association.
  - The main utility of the Bayes Net is to answer queries over what we cannot see directly from what aspects we can measure directly (assuming that there is a probabilistic association between them).
  - The work in Bayesian Networks has a lot of depth. D-separation is one aspect of it which tried to establish what is influencing what.